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Discrete Mathematics

Logic: —

⇒ Logic is the discipline that deals with the methods of reasoning.

∴ Logic is the basis of all mathematical reasoning.

∴ The rules of logic specify the meaning of mathematical statements.

Propositional Logic: —

⇒ The area of logic that deals with propositions is called the propositional logic.

Propositions: —

⇒ A proposition is a declarative statement that is either TRUE or FALSE, but not both.

✗ Statements that are not propositions include.

∴ Questions.

∴ Commands.

Key Terms:—

Propositional Variable:—

⇒ A variable that represents a proposition.

The conventional letters used for propositional variables are p, q, r, s, t .

Compound propositions:—

⇒ A proposition constructed by combining two or more propositions, using logical operators.

Logical operator:—

⇒ Operators used to combine propositions.

Truth Value:—

⇒ The truth value of a proposition is true, denoted by T , if it is a true statement. and false, denoted by F , if it is a false statement.

Truth Value $\Rightarrow$ Either True or False.

Truth Table! —

⇒ A table displaying the truth values of propositions.

Proposition Examples! —

Proposition! —

1. $3+2=32 \rightarrow F$

2. $3+2=5 \rightarrow T$

3. CSC 1204 is nobel favorite class $\rightarrow$ Either T or F

4. Every cow has four legs $\rightarrow T$

Not proposition! —

1. Bring a coffee!

2. $3+2$

3. CSC 1204 is her favorite class.

4. What time is it?

Logical Operators:

⇒ There are two types of Logical Operators.

1. Unary.

2. Binary.

① Unary:

⇒ 1. Negation ($\neg$)

② Binary:

① Conjunction ($\wedge$)

② Disjunction ($\vee$)

③ Exclusive or ($\oplus$)

④ Conditional ($\rightarrow$)

⑤ Biconditional ($\leftrightarrow$)

Logical Operators: Symbols & Usage:

Operator	Symbol	Usage	Operator	Symbol	Usage
Negation	$\neg$	NOT	Exclusive or	$\oplus$	XOR
Conjunction	$\wedge$	AND	Conditional	$\rightarrow$	if, then
Disjunction	$\vee$	OR	Biconditional	$\leftrightarrow$	iff

Propositional Logic:

✱ Negation:

- ⇒
- ① Let p be a proposition.
 - ② The negation of p is denoted by $\neg p$.
In the statement "It is not the case that p ."
 - ③ The proposition $\neg p$ is read "Not p ".
 - ④ The truth value of the negation of p , $\neg p$, is the opposite of the truth value of p .

Truth table for Negation:

p	$\neg p$
T	F
F	T

✱ Negation just turns a false proposition to true and the opposite for a true proposition.

Ex: 1:

- p : I am going to town.
- $\neg p$: I am not going to town.
- $\neg p$: It is not the case that I am going to town.

Ex: 22 —
mon $p: 15 + 7 = 20 \rightarrow F$

$\neg p: T$

Conjunction:

⇒ ① Let p and q be propositions.

①.1 The Conjunction of p and q is denoted by $p \wedge q$.
is proposition p and q .

② The Conjunction $p \wedge q$ is true when both p and q are true Otherwise is false.

③ Conjunction corresponds to English "AND"

Ex: 10 —
mon

1. Lina is curious AND clever.

Truth Table for Conjunction

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

Ex 1:
non

p : 'I am going to town'

q : 'It is going to rain'

$\therefore p \wedge q$: 'I am going to town and it is going to rain.'

Note: Both p and q must be true to $p \wedge q$ be true.

Disjunction:

1. Let p and q be propositions.

2. The Disjunction of p and q denoted by $p \vee q$, is the proposition " p or q ".

3. The Disjunction $p \vee q$ is false when both p and q are false otherwise is true.

4. Disjunction corresponds to English "OR".

Ex: 1. Abdullah is brave OR intelligent.

Truth Table for Disjunction:

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Q:1:- Let, p is True
 q is False

Construct the proposition $p \wedge q$ and $p \vee q$

$\Rightarrow$ is given:-

p is True

q is False

$\therefore$ The conjunction of proposition p and q is the proposition $p \wedge q$ is True and False

$\therefore$ The Disjunction of the proposition p and q is the proposition $p \vee q$ is True or False

Q:2:- What is the truth values of $p \wedge q$ and $p \vee q$?

p	q	$p \wedge q$	$p \vee q$
T	T	T	T
T	F	F	T
F	T	F	T
F	F	F	F

Exclusive Or:

1. Let P and Q be propositions.

2. The exclusive or of P and Q is denoted by $P \oplus Q$.

3. The Exclusive Or $P \oplus Q$ is true when exactly one of P and Q is true, Otherwise false.

Truth table for
Exclusive Or

P	Q	$P \oplus Q$
T	T	F
T	F	T
F	T	T
F	F	F

Conditional:

1. Let P and Q be propositions.

2. The Conditional Statement of P and Q is denoted by $P \rightarrow Q$, is the proposition "if P , then Q ".

3. The conditional statement $P \rightarrow Q$ is false when P is true and Q is false. Otherwise true.

4. In the conditional statement $P \rightarrow Q$, P is called hypothesis and Q is called conclusion.

5. This one is the English usage of "if, then" or "implies".

6. The connective $\rightarrow$ is called "the conditional connective".

7. A Conditional Statement is also called an implication.

Truth table for

Conditional Statement:

P	Q	$P \rightarrow Q$
T	T	T
T	F	F
F	T	T
F	F	T

Q: Why $p \rightarrow p$ is true! —

$\Rightarrow$ if $0=1$ then $0=0$.

Step 1: $0=1$ [Let]

Step 2: $1=2$ [Add 1 to both side.]

Step 3: $0=6$ [multiplied both sides by 3]

Step 4: $0=0$ [multiplied no. 1 by 3]

Step 5: $0=0$ [Add no. 3 and no. 4]

that means $0=1$ then $0=0$

Equivalent Expression of $p \rightarrow q$

$\Rightarrow$ In terms of words, the proposition

$p \rightarrow q$ also reads

- (a) if p , then q or if p, q
- (b) p implies q
- (c) p is sufficient condition for q or,
a sufficient condition for q is p
- (d) q is necessary condition for p or
a necessary condition for p is q
- (e) p only if q

Ⓐ q if p or q , if p

Ⓑ q whenever p

Ⓒ q when p

Ⓓ q unless $\neg p$

#Remember!

1. The hypothesis expresses a sufficient condition.

2. The conclusion expresses a necessary condition.

3. "but" is a logical synonym for "And"

4. "When/Whenever" means the same as "if"

5. The hypothesis is the clause following "if"

6. The conclusion is the clause following "then"

7. "Only if" clause is the "conclusion"?

8. If hypothesis, the conclusion.

Q:- Let p "Marcia Learns DM"

q "Marcia will Find a good job."

Express the statement $p \rightarrow q$ as a statement in English.

- $\Rightarrow$ "If Marcia Learns DM then she will find a good job."
- $\Rightarrow$ "Marcia Learns DM implies she will find a good job."
- $\Rightarrow$ "Marcia will Find a good job when she learns DM."
- $\Rightarrow$ "Marcia will Find a good job if she learns DM."
- $\Rightarrow$ "Marcia will Find a good job unless she does not learn DM."

Q:- Write each of these statements in the form "if p , then q " in English.

(a) It snows whenever the wind blows from the northeast.

(b) The Apple trees will bloom if it stays warm for a week.

$\Rightarrow$ (a) If the wind blows from the northeast then it snows

(b) If it stays warm for a week then the apple trees will bloom.

③ That the pistons win the championship implies that they beat the Lakers.

④ It is necessary to walk 8 miles to get to the top of Long's Peak.

⇒ If the pistons win the championship, then they beat the Lakers.

⑤ If you get to the top of Long's Peak then you must walk 8 miles.

⑥ To get tenure as a professor, it is sufficient to be world famous.

⑦ If you drive more than 400 miles, you will need to buy gasoline.

⑧ If you are world famous, then you will get tenure as a professor.

⑦ If you drive more than 400 miles, then you will need to buy gasoline.

⑨ Your guarantee is good only if you bought your CD player less than 90 days ago.

⑧ Sam will go swimming unless the water is too cold.

➡
⑧ If your guarantee is good, then you bought your CD player less than 90 days ago.

⑧ If the water is not too cold, then Sam will go swimming.

Converse, Contrapositive, and Inverse

- ① The Converse of $p \rightarrow q$ is the proposition $q \rightarrow p$
- ② The Contrapositive of $p \rightarrow q$ is the proposition $\neg q \rightarrow \neg p$
- ③ The Inverse of $p \rightarrow q$ is the proposition $\neg p \rightarrow \neg q$

Some Examples:

Converse: $p \rightarrow q \Rightarrow q \rightarrow p$

1. If it is noon, then I am hungry.

$\Rightarrow$ If I am hungry, then it is noon.

Contrapositive: $p \rightarrow q \Rightarrow \neg q \rightarrow \neg p$

1. If it is noon, then I am hungry.

$\Rightarrow$ If I am not hungry, then it is not noon.

Inverse: $p \rightarrow q \Rightarrow \neg p \rightarrow \neg q$

1. If it is noon, then I am hungry.

$\Rightarrow$ If it is not noon, then I am not hungry.

Bi Conditional:

Let, p and q be propositions.

The Bi-Conditional Statement of p and q is denoted $p \leftrightarrow q$ in the proposition " p if and only if q ".

The Bi-Conditional Statement $p \leftrightarrow q$ is true when p and q have the same truth values, otherwise false.

Bi-Conditional Statement are also called "bi-implications".

Q:- Which operator is the opposite of $\leftrightarrow$?
 $\Rightarrow \leftrightarrow$ has exactly the opposite truth table as $\oplus$.

Truth Table for Bi-Conditional:

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

Q: 10:—

Let P be the statement 'you can take the flight' and Q be the statement 'you buy a ticket'. What is the statement for $P \leftrightarrow Q$.

$\Rightarrow$ You can take the flight **if and only if** you buy a ticket.

Construct a truth table for
 $(P \oplus Q) \oplus R$

~~~~~

| $P$ | $Q$ | $R$ | $(P \oplus Q)$ | $(P \oplus Q) \oplus R$ |
|-----|-----|-----|----------------|-------------------------|
| T   | T   | T   | F              | T                       |
| T   | T   | F   | F              | F                       |
| T   | F   | T   | T              | F                       |
| T   | F   | F   | T              | T                       |
| F   | T   | T   | T              | F                       |
| F   | T   | F   | T              | T                       |
| F   | F   | T   | F              | T                       |
| F   | F   | F   | F              | F                       |



## # Precedence of Logical Operators

| Operator          | Precedence |
|-------------------|------------|
| $\neg$            | 1          |
| $\wedge$          | 2          |
| $\vee$            | 3          |
| $\rightarrow$     | 4          |
| $\leftrightarrow$ | 5          |

## # Logic & Bit Operations:-

1. bit  $\Rightarrow$  binary digit
2. Boolean Variable: either true or false
3. Can be represented by a bit.
4. Bit String:-

$\Rightarrow$  A bit string is a sequence of zero or more bits. The length of the string is the number of bits, in the string.

Example: 20

1. 101010011 is a bit string of length nine.

## #Bit Operations:-

⇒ Computer bit operations correspond to the logical connectives.

## #Table for Bit Operations:-

| P | q | $P \wedge q$ | $P \vee q$ | $P \oplus q$ |
|---|---|--------------|------------|--------------|
| 0 | 0 | 0            | 0          | 0            |
| 0 | 1 | 0            | 1          | 1            |
| 1 | 0 | 0            | 1          | 1            |
| 1 | 1 | 1            | 1          | 0            |

## #Bit String and bit operations:-

Example:  $11(+)$

The conjunction for the bit string is:-

|   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
|   | 0 | 1 | 1 | 0 | 1 | 1 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 1 |
| ∧ | 1 | 0 | 0 | 0 | 1 | 0 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 1 |
|   | 1 | 1 | 1 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 1 |

Example: 2 (x)

⇒ The conjunction for the bit string is:—

$$\begin{array}{cccccc}
 01 & 1011 & 0010 & 1001 & & \\
 \wedge & 10 & 0010 & 1111 & 1001 & \\
 \hline
 00 & 0010 & 0010 & 1001 & & 
 \end{array}$$

Example 3: (-)

⇒ The exclusive-or for the bit string is:—

$$\begin{array}{cccccc}
 01 & 1011 & 0010 & 1001 & & \\
 \oplus & 10 & 0010 & 1111 & 1001 & \\
 \hline
 11 & 1001 & 1101 & 0000 & & 
 \end{array}$$



## # Tautology

⇒ A compound proposition is called a tautology if no matter what truth values its atomic proposition have, its own truth value is T.

Example:-

1.  $P \vee \neg P$

2. The professor is either a woman or a man.

3. People either like watching TV or they don't.

Truth Table for Tautology:-

| P | $\neg P$ | $P \vee \neg P$ |
|---|----------|-----------------|
| T | F        | T               |
| F | T        | T               |

## # Contradiction:—

⇒ A compound Proposition that is always false is called a contradiction.

Examples:—

①  $p \wedge \neg p$

②  $x$  is prime and  $x$  is integer greater than 8

③ All men are good and all men are bad.

## Truth table for Contradiction:—

| $p$ | $\neg p$ | $p \wedge \neg p$ |
|-----|----------|-------------------|
| T   | F        | F                 |
| F   | T        | F                 |

## # Contingency:—

⇒ A compound proposition that is neither a tautology or a contradiction is called Contingency.

### Example:—

①  $P \rightarrow \neg P$

②  $P$

③  $\neg P$

# How to determine whether a Compound proposition is a tautology or Contradiction:—

- ① Using a truth table.
- ② Using law of equivalences.



Q8- Show that  $[\neg p \wedge (p \vee q)] \rightarrow q$  is a tautology using a truth table.

| P | q | $\neg p$ | $p \vee q$ | $\neg p \wedge (p \vee q)$ | $[\neg p \wedge (p \vee q)] \rightarrow q$ |
|---|---|----------|------------|----------------------------|--------------------------------------------|
| T | T | F        | T          | F                          | T                                          |
| T | F | F        | T          | F                          | T                                          |
| F | T | T        | T          | T                          | T                                          |
| F | F | T        | F          | F                          | T                                          |

Since, the truth table shows all the true values of compound proposition  $[\neg p \wedge (p \vee q)] \rightarrow q$  are true (T).

So it is a tautology.

H.W.  
non

- ① Determine whether  $\neg(p \wedge q) \vee p$  is a tautology or contradiction.
- ② Determine whether  $p \wedge (q \wedge \neg p)$  is a tautology or contradiction.

①

| P | q | $(p \wedge q)$ | $\neg(p \wedge q)$ | $\neg(p \wedge q) \vee p$ |
|---|---|----------------|--------------------|---------------------------|
| T | T | T              | F                  | T                         |
| T | F | F              | T                  | T                         |
| F | T | F              | T                  | T                         |
| F | F | F              | T                  | T                         |

Since, the truth table shows all the truth values of compound proposition  $\neg(p \wedge q) \vee p$  are true (T) So it's a tautology.

②

| P | q | $\neg p$ | $(q \wedge \neg p)$ | $p \wedge (q \wedge \neg p)$ |
|---|---|----------|---------------------|------------------------------|
| T | T | F        | F                   | F                            |
| T | F | F        | F                   | F                            |
| F | T | T        | T                   | F                            |
| F | F | T        | F                   | F                            |

Since the the truth table shows All the truth values of compound proposition  $p \wedge (q \wedge \neg p)$  are false (F) So, it's a contradiction.

## # Logical Equivalences:—

⇒ Compound propositions that have the same truth values in all possible cases are called logically equivalent.

### Definition:—

⇒ Compound proposition  $p$  and  $q$  are logically equivalent if  $p \leftrightarrow q$  is a tautology. denoted by  $p \equiv q$  or  $p \Leftrightarrow q$

## # How to Determine whether two Compound proposition are logically equivalent?

1. Using a Truth table.

2. Using (laws of) Logical equivalences.



Problem 1: —

⇒ Show that  $p \leftrightarrow q$  is logically equivalent to  $(p \rightarrow q) \wedge (q \rightarrow p)$

| $p$ | $q$ | $p \rightarrow q$ | $q \rightarrow p$ | $p \leftrightarrow q$ | $(p \rightarrow q) \wedge (q \rightarrow p)$ |
|-----|-----|-------------------|-------------------|-----------------------|----------------------------------------------|
| T   | T   | T                 | T                 | T                     | T                                            |
| T   | F   | F                 | T                 | F                     | F                                            |
| F   | T   | T                 | F                 | F                     | F                                            |
| F   | F   | T                 | T                 | T                     | T                                            |

Since the truth values of both of the compound propositions are same in the corresponding rows, So they are logically equivalent.

P.T.O

H.W:-

⇒ Show that  $p \vee (q \wedge r)$  and  $(p \vee q) \wedge (p \vee r)$  are logically equivalent.



| P | q | r | $q \wedge r$ | $p \vee q$ | $p \vee r$ | $p \vee (q \wedge r)$ | $(p \vee q) \wedge (p \vee r)$ |
|---|---|---|--------------|------------|------------|-----------------------|--------------------------------|
| T | T | T | T            | T          | T          | T                     | T                              |
| T | T | F | F            | T          | T          | T                     | T                              |
| T | F | T | F            | T          | T          | T                     | T                              |
| T | F | F | F            | T          | T          | T                     | T                              |
| F | T | T | T            | T          | T          | T                     | T                              |
| F | T | F | F            | T          | F          | F                     | F                              |
| F | F | T | F            | F          | T          | F                     | F                              |
| F | F | F | F            | F          | F          | F                     | F                              |

Since the truth values of both of the compound propositions are same in the corresponding rows, So, they are logically equivalent.



## # Logical Equivalences:-

| Equivalence                                                                                                                  | Name                |
|------------------------------------------------------------------------------------------------------------------------------|---------------------|
| ① $P \wedge T \equiv P$<br>② $P \vee F \equiv P$                                                                             | Identity laws       |
| ③ $P \vee T \equiv T$<br>④ $P \wedge F \equiv F$                                                                             | Domination laws     |
| ⑤ $P \vee P \equiv P$<br>⑥ $P \wedge P \equiv P$                                                                             | Idempotent laws     |
| ⑦ $\neg(\neg P) \equiv P$                                                                                                    | Double negation law |
| ⑧ $P \vee Q \equiv Q \vee P$<br>⑨ $P \wedge Q \equiv Q \wedge P$                                                             | Commutative laws    |
| ⑩ $(P \vee Q) \vee R \equiv P \vee (Q \vee R)$<br>⑪ $(P \wedge Q) \wedge R \equiv P \wedge (Q \wedge R)$                     | Associative laws    |
| ⑫ $P \vee (Q \wedge R) \equiv (P \vee Q) \wedge (P \vee R)$<br>⑬ $P \wedge (Q \vee R) \equiv (P \wedge Q) \vee (P \wedge R)$ | Distributive laws.  |
| ⑭ $\neg(P \wedge Q) \equiv \neg P \vee \neg Q$<br>⑮ $\neg(P \vee Q) \equiv \neg P \wedge \neg Q$                             | De Morgan's Laws    |
| ⑯ $P \vee (P \wedge Q) \equiv P$<br>⑰ $P \wedge (P \vee Q) \equiv P$                                                         | Absorption Laws     |
| ⑱ $P \vee \neg P = T$<br>⑲ $P \wedge \neg P = F$                                                                             | Negation Laws.      |
| 20. $P \rightarrow Q \equiv \neg P \vee Q$                                                                                   | ULE                 |



### Problem: 2

⇒ Show that  $\neg(p \rightarrow q)$  and  $p \wedge \neg q$  are logically equivalent.

$$\begin{aligned}\Rightarrow \neg(p \rightarrow q) &\equiv \neg(\neg p \vee q) && [\text{by ULE}] \\ &\equiv \neg(\neg p) \wedge \neg q && [\text{De Morgan Law}] \\ &\equiv p \wedge \neg q && [\text{Double negation Law}]\end{aligned}$$

Consequently  $\neg(p \rightarrow q)$  and  $p \wedge \neg q$  are logically equivalent.

### Problem: 3:

⇒ Show that  $\neg(p \vee (\neg p \wedge q))$  and  $\neg p \wedge \neg q$  are logically equivalent

$$\begin{aligned}\Rightarrow \neg(p \vee (\neg p \wedge q)) &\equiv \neg p \wedge \neg(\neg p \wedge q) && [\text{De Morgan Laws}] \\ &\equiv \neg p \wedge (\neg(\neg p) \vee \neg q) && [\text{De Morgan Laws}] \\ &\equiv \neg p \wedge (p \vee \neg q) && [\text{Double negation Law}] \\ &\equiv (\neg p \wedge p) \vee (\neg p \wedge \neg q) && [\text{Distributive Laws}] \\ &\equiv F \vee (\neg p \wedge \neg q) && [\text{negation Law}] \\ &\equiv (\neg p \wedge \neg q) \vee F && [\text{Commutative Law}] \\ &\equiv \neg p \wedge \neg q && [\text{Identity Law}]\end{aligned}$$

#### Problem 4

⇒ Show that  $(\neg p \wedge (p \vee q)) \rightarrow q$  is a tautology using a series of logical equivalences.

$$\Rightarrow (\neg p \wedge (p \vee q)) \rightarrow q$$

$$\Rightarrow ((\neg p \wedge p) \vee (\neg p \wedge q)) \rightarrow q \quad [\text{Distributive laws}]$$

$$\Rightarrow ((p \wedge \neg p) \vee (\neg p \wedge q)) \rightarrow q \quad [\text{Commutative laws}]$$

$$\Rightarrow (F \vee (\neg p \wedge q)) \rightarrow q \quad [\text{Negation laws}]$$

$$\Rightarrow ((\neg p \wedge q) \vee F) \rightarrow q \quad [\text{Commutative laws}]$$

$$\Rightarrow (\neg p \wedge q) \rightarrow q \quad [\text{Identity laws}]$$

$$\Rightarrow \neg(\neg p \wedge q) \vee q \quad [\text{OLE}]$$

$$\Rightarrow (\neg(\neg p) \vee \neg q) \vee q \quad [\text{De Morgan's Laws}]$$

$$\Rightarrow (p \vee \neg q) \vee q \quad [\text{Double negation Laws}]$$

$$\Rightarrow p \vee (\neg q \vee q) \quad [\text{Associative Laws}]$$

$$\Rightarrow p \vee (q \vee \neg q) \quad [\text{Commutative Laws}]$$

$$\Rightarrow p \vee T \quad [\text{Negation Laws}]$$

$$\Rightarrow T \quad [\text{Domination Laws}]$$

So,  $(\neg p \wedge (p \vee q)) \rightarrow q$  is a tautology.



Problem: 8:—

⇒ Determine whether  $(\neg p \wedge (q \rightarrow p)) \rightarrow \neg q$  is a tautology.

$$\begin{aligned} \Rightarrow (\neg p \wedge (q \rightarrow p)) \rightarrow \neg q &\equiv (\neg p \wedge (\neg q \vee p)) \rightarrow \neg q \quad [\text{ULE}] \\ &\equiv ((\neg p \wedge \neg q) \vee (\neg p \wedge p)) \rightarrow \neg q \quad [\text{Distributive laws}] \\ &\equiv ((\neg p \wedge \neg q) \vee (p \wedge \neg p)) \rightarrow \neg q \quad [\text{Commutative laws}] \\ &\equiv ((\neg p \wedge \neg q) \vee F) \rightarrow \neg q \quad [\text{Negation laws}] \\ &\equiv (\neg p \wedge \neg q) \rightarrow \neg q \quad [\text{Identity laws}] \\ &\equiv \neg(\neg p \wedge \neg q) \vee \neg q \quad [\text{ULE}] \\ &\equiv (\neg(\neg p) \vee \neg(\neg q)) \vee \neg q \quad [\text{De Morgan's Laws}] \\ &\equiv (p \vee q) \vee \neg q \quad [\text{Double negation laws}] \\ &\equiv p \vee (q \vee \neg q) \quad [\text{Associative laws}] \\ &\equiv p \vee T \quad [\text{Negation laws}] \\ &\equiv T \quad [\text{Domination laws}] \end{aligned}$$

So,  $(\neg p \wedge (q \rightarrow p)) \rightarrow \neg q$  is a tautology.



Problem 6:-

⇒ Show that  $(\neg(p \rightarrow q)) \rightarrow q$  is a tautology.

$$\Rightarrow (\neg(p \rightarrow q)) \rightarrow q \equiv (\neg(\neg p \vee q)) \rightarrow q \quad [\text{ULE}]$$

$$\equiv ((\neg \neg p) \vee (\neg q)) \rightarrow q \quad [\text{Distributive Laws}]$$

$$\equiv (p \vee (\neg q)) \rightarrow q \quad [\text{Negation Laws}]$$

$$\equiv ((\neg q) \vee p) \rightarrow q \quad [\text{Commutative Laws}]$$

$$\equiv (\neg q) \rightarrow q \quad [\text{Identity Laws}]$$

$$\equiv \neg(\neg q) \vee q \quad [\text{ULE}]$$

$$\equiv (\neg \neg q) \vee q \quad [\text{De Morgan's Laws}]$$

$$\equiv \neg \neg q \vee q \quad [\text{Associative Laws}]$$

$$\equiv \neg \neg q \vee (q \vee \neg q) \quad [\text{Commutative Laws}]$$

$$\equiv \neg \neg q \vee T \quad [\text{Negation Laws}]$$

$$\equiv T \quad [\text{Domination Laws}]$$

So,  $(\neg(p \rightarrow q)) \rightarrow q$  is a tautology.

Problem: 7:

Show that  $((p \rightarrow q) \wedge (p \rightarrow r))$  and  $(p \rightarrow (q \wedge r))$  are the logically equivalent.

$$\begin{aligned} \Rightarrow ((p \rightarrow q) \wedge (p \rightarrow r)) &\equiv (\neg p \vee q) \wedge (\neg p \vee r) && [ULE] \\ &\equiv (\neg p \vee (q \wedge r)) && [Distributive Laws] \\ &\equiv (p \rightarrow (q \wedge r)) && [ULE] \end{aligned}$$

So,  $((p \rightarrow q) \wedge (p \rightarrow r))$  and  $(p \rightarrow (q \wedge r))$  are logically equivalent.

## Lecture 4: —

### # Predicates —

⇒ A property that the subject of the statement can have.

Example: —

" $x > 3$ "

where,  $x$  = Variable

$> 3$  = is a predicate.

① We can denote the statement " $x$  is greater than 3" by  $P(x)$ . The statement  $P(x)$  is also said to be the value of the propositional function  $P$  at  $x$ .

where,

$P$  = the predicate " $x$  is greater than"

$x$  = the variable.

Notes: —

⇒ Once a value has been assigned to the variable  $x$ , the statement  $P(x)$  becomes a proposition and has a truth value.

② A predicate, or a propositional function, is a function that takes some variables as argument and returns True or False.



③ A predicate is symbolized by Capital Letter (P) and the Variable (x) by small letter (x)

③  $P(x)$  is also called the propositional function, because each value of  $x$  produces a proposition  $P(x)$  is either true or false.

Example 1:

Let  $P(x)$  denote the statement " $x > 3$ "

What are the truth values of  $P(4)$  and  $P(2)$ ?

⇒ is given:

$$P(x) = x > 3$$

$$\therefore P(4) = 4 > 3 \text{ is true}$$

$$\therefore P(2) = 2 > 3 \text{ is false.}$$

Example 2:

Let,  $A(x)$  "Computer  $x$  is under attack by an intruder". Suppose that of the computers on campus only  $c_1$  and  $c_7$  are currently under attack by intruders. What are the truth values of  $A(c_1)$ ,  $A(c_3)$ ,  $A(c_7)$ ?

⇒ ①  $A(c_1)$ : "Computer  $c_1$  is under attack by an intruder" = True

②  $A(c_2)$ : "Computer  $c_2$  is under attack by an intruder" = True.

③  $A(c_3)$ : "Computer  $c_3$  is under attack by an intruder" = False.

Because  $c_3$  is not in the list of computers that are attacked by intruders.

### # Types of Predicates:

⇒ There are three types of Predicates

1. Unary → 1 Variable

2. Binary → 2 Variable

3. Ternary → 3 Variable



## # Multivariable

### Predicates:-

⇒ Predicates that have more than one Variable is called multivariable Predicates.

Ex:-

$$Q(x, y) : "x = y + 3"$$

Note:-

$x$  = Variables

$y$  = Variables

$Q$  = predicate

Note:- When values are assigned to the Variables  $x$  and  $y$ , the statement  $Q(x, y)$  has a truth value.

Problem:-

⇒ Let,  $Q(x, y)$  denote the statement " $x = y + 3$ ". What are the truth values of the propositions  $Q(1, 2)$  and  $Q(3, 0)$ ?

⇒ is given:-

the statement  $x = y + 3$

∴  $Q(1, 2)$  ∵  $1 = 2 + 3$  is false

Similarly  $Q(3, 0)$  ∵  $3 = 0 + 3$   
 $3 = 3$  is true.



## # Quantifiers:-

⇒ There are two types of quantification

1. Universal Quantification

2. Existential Quantification.

### Universal Quantification:-

⇒ A predicate is true for every element in the domain.  $\forall$  is called the Universal quantifier. " $\forall$ " reads "for All".

### Existential Quantification:-

⇒ There is one or more elements in the Domain for which a predicate is true.  $\exists$  is called Existential Quantifier. " $\exists$ " reads "there Exists".

### Domain/Domain of discourse/universe of discourse:-

→ The values a variable in a propositional function may take.

## # The Universal Quantifier:-

⇒ The Universal quantification of  $P(x)$  is the statement " $P(x)$  for all the values of  $x$  in the domain".

- ① The Universal Quantification of  $P(x)$  is denoted by  $\forall x P(x)$ .
- ② We read  $\forall x P(x)$  as "for all  $x$ ,  $P(x)$ " or "for every  $x$ ,  $P(x)$ ".
- ③ An element for which  $P(x)$  is false is called a counterexample of  $\forall x P(x)$ .
- ④ " $\forall x P(x)$ " is true when every instance of  $x$  makes  $P(x)$  true.
- ⑤ Like taking conjunction over the entire universe

$$\forall x P(x) \equiv P(x_1) \wedge P(x_2) \wedge P(x_3) \wedge \dots \wedge P(x_n)$$

Problem:- 8

⇒ Let  $P(x)$  be the statement " $x+1 > x$ " what is the truth value of the quantification  $\forall x P(x)$ , where the domain consists of all real numbers.



⇒ is given:-

the statement " $x+1 > x$ " =  $P(x)$   
and domain All real numbers

Now, If we add 1 to any real number  $x$ ,  
that number is always bigger than  $x$ .

then,  $P(x)$  is true for all real numbers  $x$ .

So, the truth value of the quantification  $\forall x P(x)$  is True.

Problem 9:-

Let  $Q(x)$  be the statement " $x < 2$ ". What is the  
Truth value of the quantification  $\forall x Q(x)$ , where  
the universe of discourse consists of all  
Integers?

⇒ is given:-

The Statement  $Q(x) \equiv x < 2$

Domain = All Integers.

Now:-

$Q(3) \equiv 3 < 2$  is false

So,

$Q(x)$  is not true for every integers numbers.  
That  $x=3$  is a Counter example for the  
Statement  $\forall x Q(x)$

So, The truth value of the  $\forall x Q(x)$  is False.



Problem 10:-

⇒ Let  $P(x)$  be the statement " $x^2 > 0$ ". What is the truth value of the quantification  $\forall x P(x)$ , where Domain consists of all integers.

⇒ is given:-

the Statement  $P(x) \equiv x^2 > 0$

Domain = All integers.

Now:-

$P(0) \equiv 0^2 > 0$ ,  $0 > 0$  is false.

So,  $P(x)$  Statement is not true for all Integer numbers.

That  $x=0$  is a Counterexample for the statement  $\forall x P(x)$

So, The truth value of the  $\forall x P(x)$  is false.

Problem: 11:-

⇒ What is the truth value of  $\forall x P(x)$ , where  $P(x)$  is the statement " $x^2 \geq 10$ " and the domain consists of the positive integers not exceeding 4.

⇒ is given:-

The statement  $P(x) \equiv x^2 \geq 10$

Domain = 1, 2, 3, 4

Now:-

$P(4) \equiv 4^2 \geq 10$  is false.

Then:-

$P(x)$  is not true for. Consists Domain

So,

The truth value of  $\forall x P(x)$  is false.

Problem 12:

⇒ What is the truth value of  $\forall x P(x)$ , where  $P(x)$  is the statement " $x^2 \leq 10$ ", and the Domain consists of the positive integers less than 4?

⇒ is given:

The Statement  $P(x) \equiv x^2 \leq 10$

Domain = 1, 2, 3

Now,  $P(1) \equiv 1^2 \leq 10$  is true

$P(2) \equiv 2^2 \leq 10$  is true

$P(3) \equiv 3^2 \leq 10$  is true

Then:-  $P(x)$  is true for the consists Domain

So, The truth value of the  $\forall x P(x)$  is true.



## # The Existential Quantifier:-

⇒ The existential quantifier of  $P(x)$  is the proposition  
"There exists an element  $x$  in the domain such that  $P(x)$ ".

① We denote the existential Quantification of  $P(x)$  by  $\exists x P(x)$

② Existential quantification is read as  $\exists x P(x)$ :-

- (i) There is an  $x$  such that  $P(x)$
- (ii) There is at least one  $x$  such that  $P(x)$
- (iii) For some  $x$   $P(x)$

④  $\exists x P(x)$  is true when an instance can be found which when plugged in for  $x$ , makes  $P(x)$  true.

⑤ Like taking disjunction over the entire domain

$$\exists x P(x) \equiv P(x_1) \vee P(x_2) \vee P(x_3) \vee \dots \vee P(x_n)$$



Problem 14:

Let  $P(x)$  denote the statement " $x > 3$ ". What is the truth value of the quantification  $\exists x P(x)$ , where the domain consists of all real numbers.

In given:

The Statement  $P(x) \equiv "x > 3"$

Domain = All real numbers.

Now,

$P(4) \equiv 4 > 3$  is true

Then,

$P(x)$  is true for some real numbers

So, The truth value of  $\exists x P(x)$  is true

Problem 15:

Let  $Q(x)$  denote the statement " $x = x + 1$ " what is the truth value of the quantification  $\exists x Q(x)$  where Domain consists all the real numbers?

In given:

The Statement  $Q(x) \equiv "x = x + 1"$

Domain = All real numbers.

Now: If we add 1 to any real number  $x$ , that number will never be equal to  $x$ , It will always be bigger than  $x$ .

Then:-  $Q(x)$  is false for All real numbers

So, The truth value of  $\exists x P(x)$  is false.

Problem: 16:-

⇒ What is the truth value of  $\exists x P(x)$ , where  $P(x)$  is the statement " $x^2 > 10$ " and the universe of discourse of the positive integers not exceeding 4?

⇒ is given:-

The Statement  $P(x) \equiv x^2 > 10$

Domains = 1, 2, 3, 4

Now:-  $P(4) \equiv 4^2 > 10$  is true

Then:-  $P(x)$  is true for some positive integers/Domain

So: The truth value of  $\exists x P(x)$  is true.

### #class work

Problem: 1

⇒ Let  $P(x)$  denote the statement " $x > 0$ ". What is the truth value of the quantification  $\exists x P(x)$ , where the Domain consists of integers?

⇒ is given:-

The Statement  $P(x) \equiv x > 0$

Domain = All integers.

Now:-  $P(4) \equiv 4 > 0$  is true.

Then:-  $P(x)$  is true for some integers

We know:-

$\exists x P(x)$  is the same as the disjunction

So: The truth value of  $\exists x P(x)$  is true.



### Problem 2:-

Let  $P(n)$  denote the statement " $n > 0$ ". What is the truth value of the quantification  $\forall n P(n)$ , where domain consists of non-negative integers?

is given:-

The Statement  $P(n) \equiv n > 0$

Domains = non-negative integers.

Now:-  $P(0) \equiv 0 > 0$  is false.

Then:-  $P(n)$  is false for one non-negative integer.

So, The truth value of  $\forall n P(n)$  is false.

### Problem 3:-

Let  $P(n)$  denote the statement " $n > 0$ ". What is the truth value of the quantification  $\exists n P(n)$ , where the Domain consists the negative integers.

is given:-

The Statement  $P(n) \equiv n > 0$

Domain = negative integers.

Now,

If we use any negative integer in the Statement, there will never be greater than zero.

Then:-  $P(n)$  is false for all negative integers

So:- The truth value of the  $\exists n P(n)$  is false.

#### Problem: 4

⇒ Let  $P(n)$  denote the statement " $n < 2$ ". What is the truth value of the quantification  $\exists n P(n)$ , where the Domain consists all the prime numbers.

⇒ is given:

The statement  $P(n) \equiv n < 2$

Domain = all prime numbers.

we know:-

$\exists n P(n)$  is the same as disjunction

so,  $P(2) \equiv 2 < 2$  is false

$P(3) \equiv 3 < 2$  is false

we also know that the smallest prime number is 2

Then:-  $P(n)$  is false for all prime numbers.

so, The truth value of  $\exists n P(n)$  is False.

#### Problem: 5:

⇒ Let  $P(n)$  denote the statement " $n \leq 2$ ". What is the truth value of the quantification  $\exists n P(n)$ , where the Domain consists all the prime numbers.

⇒ is given:

$P(n) \equiv n \leq 2$

Domain = all prime number.



We know:

$\exists x P(x)$  is the same as disjunction.

Now:

$P(2) \equiv 2 \leq 2$  is true

Then:-

$P(x)$  is true for one prime number

So,

The truth value of  $\exists x P(x)$  is true.

### # Precedence of Quantifiers:-

- ① The Universal and existential quantifiers have higher precedence for all logical operators from propositional calculus.

Ex:

i)  $\forall x P(x) \vee \phi(x)$  is the disjunction of  $\forall x P(x)$  and  $\phi(x)$ .

ii)

it means  $\forall x (P(x) \vee \phi(x))$  rather than  $\forall x (P(x) \vee \phi(x))$

## # Negating Quantified expressions:—

### De Morgan's Laws for Quantifiers:—

⇒ The rules for negation for quantifiers are called De Morgan's laws for quantifiers.

⇒ ① Negation of Conjunctions:—

$$\neg(P_1 \wedge P_2 \wedge P_3 \wedge \dots \wedge P_n) \equiv (\neg P_1 \vee \neg P_2 \vee \dots \vee \neg P_n)$$

② Negation of Disjunctions:—

$$\neg(P_1 \vee P_2 \vee P_3 \vee \dots \vee P_n) \equiv (\neg P_1 \wedge \neg P_2 \wedge \dots \wedge \neg P_n)$$

⇒ Since the Quantifiers are the same as taking a bunch of (AND) or (OR) we have:—

① Universal Negation:—

$$\neg \forall x P(x) \equiv \exists x \neg P(x)$$

② Existential Negation:—

$$\neg \exists x P(x) \equiv \forall x \neg P(x)$$



# Translate from English into logical expression:—

Problem: 23:—

⇒ Express the statement <sup>"Every student"</sup> "in the class has studied calculus" using predicate and quantifiers.

⇒ We write the statement:—

"For every student in the class, that student has studied calculus."

Introduce variable  $x$ :—

for every student  $x$  in the class,  $x$  has studied calculus.

Now, introduce the predicate  $C(x)$ :—

$C(x)$ : " $x$  has studied calculus"

Now, introduce the quantifiers:—

$\forall x C(x)$

Problem:-

⇒ Express the statement "Someone in your school has studied DM" using predicates and quantifiers. Let the domain consists of all people.

⇒ Let

$S(x)$  =  $x$  is in your school. and  
 $C(x)$  =  $x$  has studied DM

$$\exists x (S(x) \wedge C(x))$$

Note:-

if domain consists of the "students in your school", then we can write  $\exists x C(x)$

# Exercise

⇒ Let  $P(x)$  be the statement " $x$  can speak Russian" and let  $Q(x)$  be the statement " $x$  knows the computer language C++". Express each of these sentences in terms of  $P(x)$ ,  $Q(x)$ , quantifiers, and logical connectives. The domain for quantifiers consists of all students at your school.

① There is a student at your school who can speak Russian who knows C++.



⑥ There is a student at your school who can speak Russian but who does not know c++

⑦ Every student in your school either can speak Russian or knows c++

⑧ No student at your school can speak Russian or knows c++.

⇒ Let:- in given:-

$P(x)$  =  $x$  can speak Russian

$Q(x)$  =  $x$  knows the computer language c++

①

$\exists x (P(x) \wedge Q(x)) \rightarrow$  Domain = a student at your school

②

$\exists x (P(x) \wedge \neg Q(x)) \rightarrow$  Domain = a student at your school.

③

$\forall x (P(x) \vee Q(x)) \rightarrow$  Domain = Every student in your school.

④

$\forall x \neg (P(x) \vee Q(x)) \rightarrow$  Domain = No student in your school

### Exercise 25:

→ Translate each of the statements into logical expressions. using predicates, quantifiers, and logical connectives. Let the Domain be all people.

- (a) No one is perfect
- (b) Not every one is perfect
- (c) All your friends are perfect
- (d) At least one of your friends is perfect
- (e) Every one your friends is perfect
- (f) Not everybody is your friend or someone is not perfect

⇒

Let:-

(a)

$P(x) \equiv "x \text{ is perfect}"$

$\therefore \forall x \neg P(x)$

(b)

Let:-

$P(x) \equiv "x \text{ is perfect}"$

$\neg \forall x P(x)$



⑧

Let:-

$P(x)$ : "x is perfect"

$F(x)$ : "x is your friend"

$$\forall x (F(x) \rightarrow P(x))$$

⑨

Let:-

$P(x)$ : "x is perfect"

$F(x)$ : "x is your friend"

$$\exists x (F(x) \wedge P(x))$$

⑩

$P(x)$ : "x is perfect"

$F(x)$ : "x is your friend"

$$\forall x (F(x) \wedge P(x))$$

⑪

$P(x)$ : "x is perfect"

$F(x)$ : "x is your friend"

$$(\neg \forall x F(x)) \vee (\exists x \neg P(x))$$

## Lecture : 5

### #Functions:-

$\Rightarrow$  Let  $A$  and  $B$  be nonempty Sets.

A function  $f$  from  $A$  to  $B$  is a set of ordered pairs  $f \subseteq A \times B$  with the property that for each element ' $a$ ' in  $A$  there is a unique element ' $b$ ' in  $B$  such that  $(a, b) \in f$

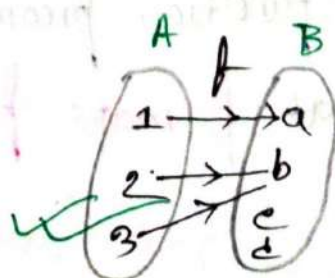
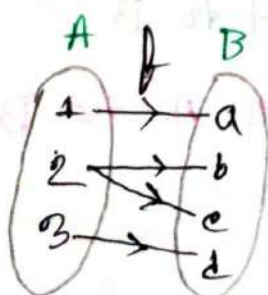
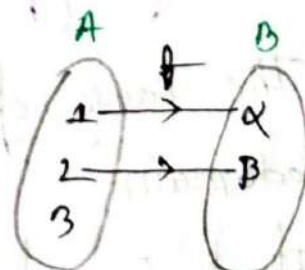
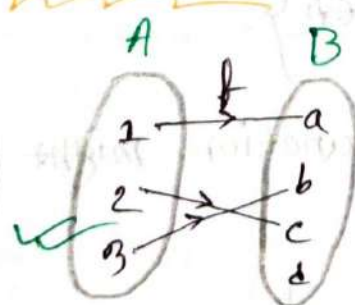
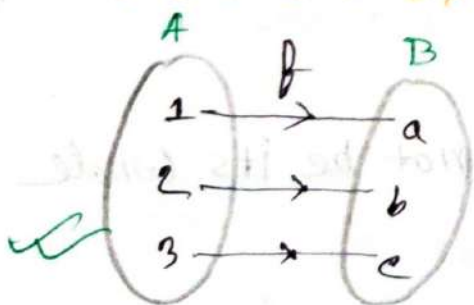
① if  $f$  is function from  $A$  to  $B$ ,

Then, we can write  $f: A \rightarrow B$

② Function are sometime called mappings or transformations.

③ A relation  $A$  to  $B$  is just subset of  $A \times B$

### Some example of function:-





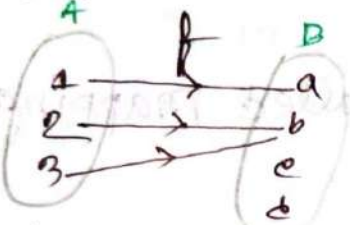
## # Some Function Terminology:-

$\Rightarrow$  If  $f$  is a function from  $A$  to  $B$ , we say that  $A$  is the domain of  $f$  and  $B$  is the codomain of  $f$ .

① If  $f(a) = b$ ,

$a$  is the preimage of  $b$  and  $b$  is the image of  $a$ .

② Range of  $f$  is the set of all images of elements of Domain.



Domain

$$A = \{1, 2, 3\}$$

Codomain

$$B = \{a, b, c, d\}$$

$$\text{Range} = \{a, b\}$$

$$\text{Range} = \{f(x) : x \in A\}$$

③ The Range of a function might not be its whole Codomain.

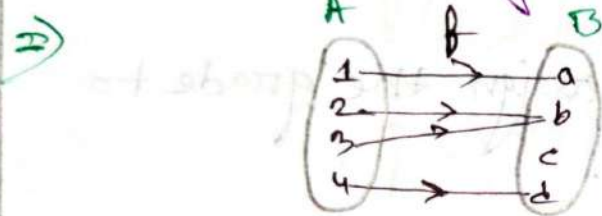
④ If  $f$  is a function from  $A$  to  $B$ , we say that  $f$  maps from  $A$  to  $B$ .

Example:-

Let,  $A = \{1, 2, 3, 4\}$  and  $B = \{a, b, c, d\}$  and let  $f = \{(1, a), (2, b), (3, b), (4, d)\}$

If  $f$  is a function from  $A$  to  $B$  then

- ① Find the Domain?
- ② Find the CoDomain?
- ③ if  $(a, b) \in f$  Find the Image and preimage.
- ④ Find the Range.



① The Domain  $f: A = \{1, 2, 3, 4\}$

② Codomain  $f: B = \{a, b, c, d\}$

③ is given:-

$(a, b) \in f$ , then  $f(a) = b$

- ①  $b$  is called the image of  $a \in \{a, b, c, d\}$
- ②  $a$  is called pre-image of  $b \in \{1, 2, 3, 4\}$

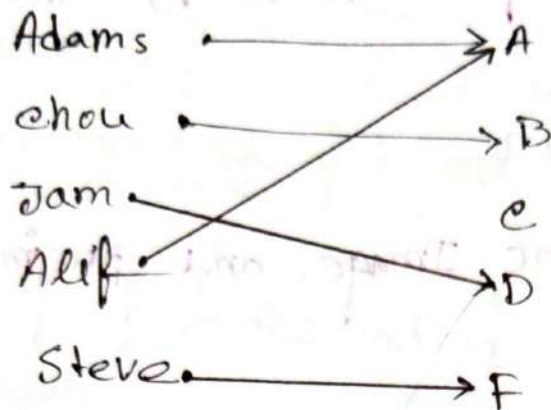
④ we know:-

Range  $= \{f(x) : x \in A\}$   
 $f(A) = \{a, b, d\}$



Problem:-

⇒ What are the domain, codomain and range of the function that assigns grades to students of class as follows.?



⇒ Let,  $f$  is a function that assign the grade to a student of DM class.

∴ The Domain  $f = \{ \text{Adams, Chou, Jam, Alif, Steve} \}$

∴ The Codomain  $f = \{ A, B, C, D, F \}$

∴ The Range  $f = \{ A, B, D, F \}$

Because each grade except C is assigned to some student

### Problem 2

Let,  $R$  be the Relation consisting of ordered pairs  $(\text{Sami}, 22), (\text{Burr}, 21), (\text{Mohsin}, 22), (\text{Alif}, 10), (\text{Jam}, 24), (\text{Farrah}, 22)$ , where each pair consists of a graduate student and the age of this student, what is the function that this relation determines?

⇒ This relation defines the function  $f$ ,

where, with  $f(\text{Sami}) = 22, f(\text{Burr}) = 21,$

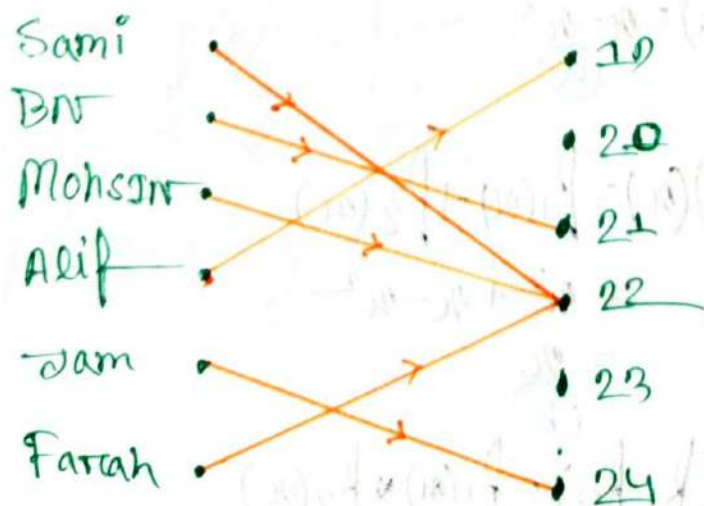
$f(\text{Mohsin}) = 22, f(\text{Alif}) = 10,$

$f(\text{Jam}) = 24, f(\text{Farrah}) = 22$

So, the Domain  $f = \{ \text{Sami, Burr, Mohsin, Alif, Jam, Farrah} \}$

the Codomain  $f = \text{Set of positive integers or}$   
 $\{ 22, 21, 22, 10, 24, 22 \}$

Range  $f = \{ 22, 21, 10, 24 \}$





## # Function:-

### Definition: 3

$\Rightarrow$  Let  $f_1$  and  $f_2$  be function from  $A$  to  $R$ .  
Then  $f_1 + f_2$  and  $f_1 f_2$  are also functions  
from  $A$  to  $R$  defined by

$$(f_1 + f_2)(x) = f_1(x) + f_2(x)$$

$$(f_1 f_2)(x) = f_1(x) \cdot f_2(x)$$

### Problem: 6:-

$\Rightarrow$  Let  $f_1$  and  $f_2$  be functions from  $R$  to  $R$   
Such that  $f_1(x) = x^2$  and  $f_2(x) = x - x^2$ . What are  
the function of  $f_1 + f_2$  and  $f_1 f_2$

$\Rightarrow$  is given:-

$$f_1(x) = x^2$$

$$f_2(x) = x - x^2 \quad \textcircled{a}$$

we know:-

$$(f_1 + f_2)(x) = f_1(x) + f_2(x)$$

$$= x^2 + x - x^2$$

$$= x$$

$\textcircled{b}$

we know:-

$$(f_1 \cdot f_2)(x) = f_1(x) \cdot f_2(x)$$

$$= x^2 \cdot (x - x^2)$$

$$= x^3 - x^4$$

## # Functions:-

### Definition:-

⇒ Let  $f$  be a function from the set  $A$  to the set  $B$ , and let  $S$  is a subset of  $A$ . The image of  $S$  under the function  $f$  is the subset of  $B$  that consists of the images of the elements of  $S$ . We denote the image of  $S$  by  $f(S)$ .

$$\therefore f(S) = \{x \mid \exists s \in S (x = f(s))\}$$

### Problem:-

⇒ Let,  $A = \{a, b, c, d, e\}$  and  $B = \{1, 2, 3, 4\}$  with  $f(a) = 2$ ,  $f(b) = 1$ ,  $f(c) = 4$ ,  $f(d) = 1$ ,  $f(e) = 1$ .

What the image of the subset  $S = \{b, c, d\}$ ?

⇒ in given:-

$$A = \{a, b, c, d, e\}$$

$$B = \{1, 2, 3, 4\}$$

$$f(a) = 2, f(b) = 1, f(c) = 4, f(d) = 1, f(e) = 1$$

$\therefore$  The image of the Subset of  $S = \{b, c, d\}$  is the set  $f(S) = \{1, 4\}$ .



## # Types of Function:-

⇒ There are four types

- (i) one-to-one (Injective)
- (ii) many-to-one
- (iii) Into
- (iv) Onto (Surjective)

## Other types:-

- (v) One-one into
- (vi) one-one onto (Bijective)
- (vii) many-one into
- (viii) many-one onto

## # One-to-One Function: (Injective)

⇒ A function  $f: A \rightarrow B$  is called one-to-one function, if distinct element of  $A$  are mapped into distinct elements of  $B$ . In other words  $f$  is one-to-one if

$$x_1 \neq x_2 \Rightarrow f(x_1) \neq f(x_2)$$

or equivalently:—

$$x_1 = x_2 \Rightarrow f(x_1) = f(x_2)$$

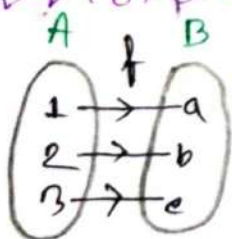
① We can express that  $f$  is one-to-one using quantifiers as

$\forall a \forall b (f(a) = f(b) \rightarrow a = b)$ , or equivalently,

~~$\forall a \forall b (f(a) \neq f(b) \rightarrow a \neq b)$~~

$\forall a \forall b (a \neq b \rightarrow f(a) \neq f(b))$  where the universe of discourse is the domain of the function  $f$

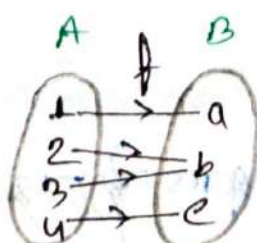
Graphically:—



One-to-one

$$|A| = 3, |B| = 3$$

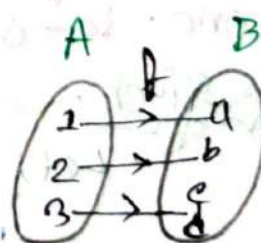
$$|A| = |B|$$



Not one-to-one

$$|A| = 4, |B| = 3$$

$$|A| > |B|$$



One-to-one

$$|A| < |B|$$

$$|A| \neq 3, |B| = 4$$

So, the condition of one-to-one:—  
 $|A| \leq |B|$



Problem 8:

Q Determine whether the function  $f$  from  $\{a, b, c, d\}$  to  $\{1, 2, 3, 4, 5\}$  with  $f(a)=4, f(b)=5, f(c)=1, f(d)=3$  is one to one.

is given—

Let,  $A = \{a, b, c, d\}$

$B = \{1, 2, 3, 4, 5\}$

$f(a)=4, f(b)=5, f(c)=1, f(d)=3$

Since Every element of Domain has a distinct Image.

So, The function is a one-to-one function.

Problem 9:

Q Determine whether the function  $f(x) = x^2$  from the set of integers to the set of integers is one-to-one.

is given,

Let,  $A = \text{All integers}$

$B = \text{All integers}$

$f(x) = x^2$

Now,

$f(1) = 1^2 = 1$

$f(-1) = (-1)^2 = 1$

Since, Every element of Domain has not distinct image

So, The function  $f(x) = x^2$  is not one-to-one function

# class work:-

⇒ Determine the whether the function  $f(x) = x^2$  from the set of positive integers to the set of positive integers is one-to-one function.

⇒ In given:-

the function  $f(x) = x^2$

Let,

A = All positive integers

B = All positive integers.

Now,

$$f(1) = 1^2 = 1$$

$$f(2) = 2^2 = 4$$

$$f(3) = 3^2 = 9 \dots$$

Similarly, Every element of Domain has a distinct Image.

So, The function  $f(x) = x^2$  is a one-to-one function



Problem 10:-

⇒ Determine whether the function  $f(x) = x+1$  from the set of real numbers to the set of the real numbers is one-to-one.

⇒ in given:-

Let,  $f(x) = x+1$

Set,  $A = \text{Real numbers}$

$B = \text{Real numbers.}$

Let,

now,

$$f(-1) = -1+1 = 0$$

$$f(0) = 0+1 = 1$$

$$f(1) = 1+1 = 2$$

Similarly, For any real number  $x$ , there is a distinct image, just 1 bigger than  $x$ .

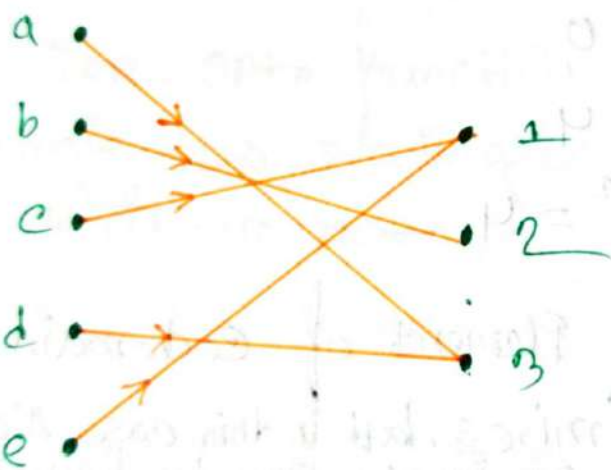
So:- The function  $f(x) = x+1$  is a one-to-one function.

## # Onto function (Surjective)

⇒ A function  $f: A \rightarrow B$  is onto or surjective, if every element of  $B$  is the image of some element of  $A$ , that is if  $B = \text{range of } f$ .

① A function is Onto if every element of codomain has preimage(s).

Graphically:—





Problem: 12

⇒ In the function  $f(x) = x^2$  from the set of integers to the set of integers onto

is given: —

$$f(x) = x^2$$

Let, Domain = Set of integers

Codomain = Set of integers.

Let,

Now,

$$f(-1) = (-1)^2 = 1$$

$$f(1) = (1)^2 = 1$$

$$f(0) = 0^2 = 0$$

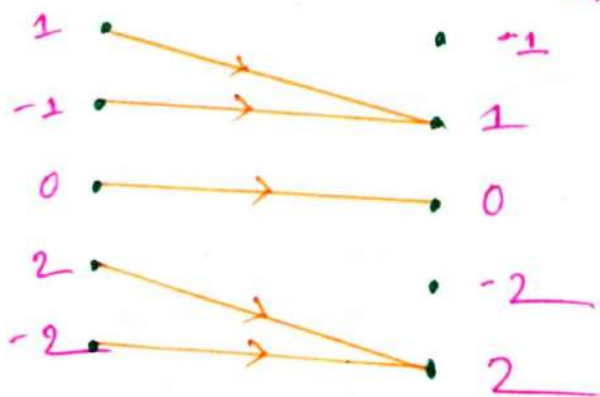
$$f(2) = 2^2 = 4$$

$$f(-2) = (-2)^2 = 4$$

we need

Similarity: — Every Element of codomain has a Premises, but in this case No negative Codomain preimage can be found.

So, the function  $f(x) = x^2$  is not Onto function.



Problem: \*

⇒ In the function  $f(x) = x^2$  from the set of positive integers to the set of p. integers onto?

⇒ In given:—

$$f(x) = x^2$$

Domain = positive integers

Let, Codomain = positive integers

Now:—

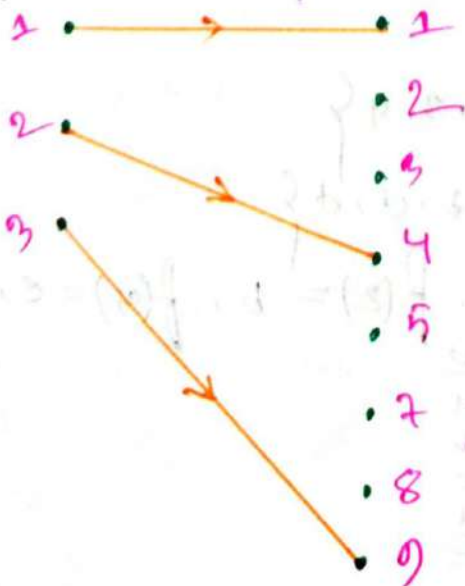
$$f(1) = 1^2 = 1$$

$$f(2) = 2^2 = 4$$

$$f(3) = 3^2 = 9$$

We know:— For onto function every element of codomain has preimages. But in this function  $f(x) = x^2$ , it is not possible to get the preimage of all the codomain.

So, the function  $f(x) = x^2$  is not-onto function



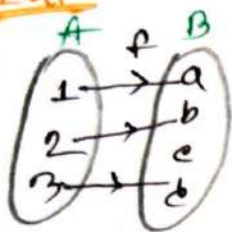


## # One-to-One Correspondence

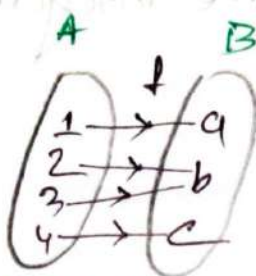
(Bijection!)

$\Rightarrow$  A function  $f$  is a one-to-one correspondence or Bijection if it is both one-to-one and onto.

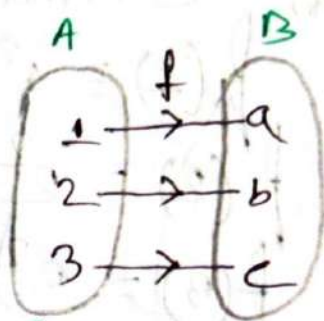
Ex:-



✓ one-to-one  
✗ not onto  
✗ not one-to-one



✓ onto



✓ one-to-one  
✓ onto  
(Bijection)

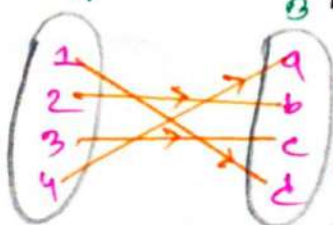
Problem: \*

Let,  $f$  be the function from  $A$  to  $B$  where  
 $A = \{1, 2, 3, 4\}$ ,  $B = \{a, b, c, d\}$ , with  $f(1) = d$ ,  $f(2) = b$ ,  
 $f(3) = c$ ,  $f(4) = a$ , then  $f$  is bijective function  
 $\Rightarrow$  is given:-

$$A = \{1, 2, 3, 4\}$$

$$B = \{a, b, c, d\}$$

$$f(1) = d, f(2) = b, f(3) = c, f(4) = a$$



check one-to-one:

we know:

$\Rightarrow$  The function is one-to-one if every element of Domain has a Distinct image.

So, for this function  $f: A \rightarrow B$  is a one-to-one function.

check onto:

we know:

$\Rightarrow$  For Onto function every element of codomain has preimages.

Since, every element of codomain has preimages  
So, it's an onto function.

Hence,  $f$  is a bijective function.



## Lecture: 6

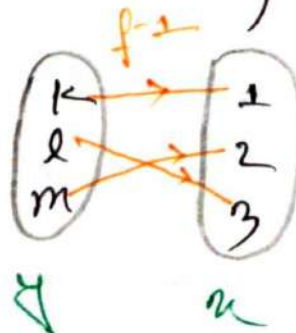
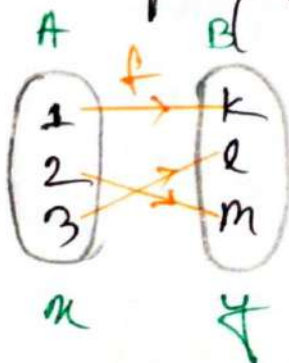
### # Inverse Function:-

$\Rightarrow$  If  $f: A \rightarrow B$  is a Bijective function, Then function defined by  $g: B \rightarrow A$  is called inverse function of  $f$ . The inverse  $g$  of  $f$  denoted by  $f^{-1}$ .

- (i)  $f(a) = b, f^{-1}(b) = a$
- (ii) A one-to-one correspondence is called invertible because we can define an inverse of this function.
- (iii) A function is not invertible if it is not a Bijective function.

### Problem: Example:-

$$X = \{1, 2, 3\}, Y = \{k, l, m\}$$
$$f: X \rightarrow Y \text{ such that}$$
$$f = \{(1, k), (2, m), (3, l)\}$$



Problem: 16: —

Let  $f$  be the function from  $\{a, b, c\}$  to  $\{1, 2, 3\}$  such that  $f(a) = 2$ ,  $f(b) = 3$ ,  $f(c) = 1$  in  $f$  invertible and if it is, what is its inverse?

⇒ The function  $f$  is invertible because it is a Bijective function.

is given:—

$$\text{Let, } A = \{a, b, c\}$$

$$B = \{1, 2, 3\}$$

$f: A \rightarrow B$  such that

$$f = \{(a, 2), (b, 3), (c, 1)\}$$

So, the inverse function  $f^{-1}$  is reverse the correspondence given by  $f$ .

So,  $f^{-1} = \{(2, a), (3, b), (1, c)\}$

$$f^{-1}(1) = c, f^{-1}(2) = a, f^{-1}(3) = b$$



Example: 121 —

⇒ Let  $f: \mathbb{Z} \rightarrow \mathbb{Z}$  be such that  $f(x) = x+1$   
is  $f$  invertible? and if it is, what is its inverse?

⇒ The function  $f$  is one-to-one because every element of domain has a dist int image, just one bigger than that element.

⇒ The function  $f$  is a onto because every element of codomain has a preimage, just 1 smaller than that element.

So, the function  $f: \mathbb{Z} \rightarrow \mathbb{Z}$  is a Bijective function  
Then the function  $f$  is invertible function!

# Find the inverse! —

⇒ To reverse the Correspondence,  
Let  $y$  is the image of  $x$ ,

So

$$y = x+1$$
$$x = y-1$$

That means  $y-1$  is the unique element of  $\mathbb{Z}$  that is sent to  $y$  by  $f$

$$\therefore f^{-1}(y) = y-1$$

Problem, 18.

Let,  $f$  be the function from  $\mathbb{R}$  to  $\mathbb{R}$  with  $f(x) = x^2$   
is  $f$  invertible?

$\Rightarrow$  in given:-

$$f(x) = x^2$$
$$f: \mathbb{R} \rightarrow \mathbb{R}$$

Now:-  $f(-1) = (-1)^2 = 1$

$$f(1) = (1)^2 = 1$$

$$f(-2) = (-2)^2 = 4$$

$$f(2) = (2)^2 = 4 \dots \dots$$

Since, Every element of Domain has not distinct image  
So, it's not a one-to-one function.

That means:- it's not a Bijective function.

Then:-  $f$  is not invertible.



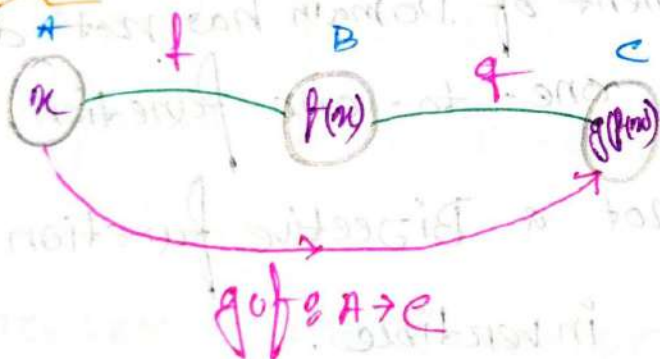
## # Compositions of Function:-

⇒ Let,  $g: A \rightarrow B$  and  $f: B \rightarrow C$ , The composition of function  $f$  and  $g$  denoted by  $f \circ g$  (f o g) is defined by  $(f \circ g)(a) = f(g(a))$

①  $f \circ g$  and  $g \circ f$  are not equal

② The composition  $f \circ g$  can NOT be defined Unless the range of  $g$  is a subset of the domain of  $f$

Example:-



Problem 2:-

⇒ Let  $f$  and  $g$  be the function from the set of integers to the set of integers defined by  $f(x) = 2x + 3$  and  $g(x) = 3x + 2$ . Find the composite function of  $f$  and  $g$ .

⇒ is given:—

$$f(x) = 2x + 3$$

$$g(x) = 3x + 2$$

① The composition of the function  $f$  and  $g$  is

$$(f \circ g)(x) = f(g(x))$$

$$= f(3x + 2)$$

$$= 2(3x + 2) + 3$$

$$= 6x + 4 + 3$$

$$= 6x + 7$$

② The composition of the function  $g$  and  $f$  is:—

$$(g \circ f)(x) = g(f(x))$$

$$= g(2x + 3)$$

$$= 3(2x + 3) + 2$$

$$= 6x + 9 + 2$$

$$= 6x + 11$$



## # Floor Function:-

⇒ The greatest integer that is less than or equal to the  $x$ .

① The value of the floor function at  $x$  is denoted by  $\lfloor x \rfloor$

Example:-

①  $\lfloor 2.3 \rfloor = 2$

②  $\lfloor 2 \rfloor = 2$

③  $\lfloor 0.5 \rfloor = 0$

④  $\lfloor -3.5 \rfloor = -4$

## # Ceiling Function:-

⇒ The greatest integer that is greater than or equal to  $x$

① The value of the ceiling function at  $x$  is denoted by  $\lceil x \rceil$

Example:-

①  $\lceil 2.3 \rceil = 3$

②  $\lceil 2 \rceil = 2$

③  $\lceil 0.5 \rceil = 1$

④  $\lceil -3.5 \rceil = -3$

Problem: 1:-

How many bytes are required to encode 600 kilobits of data? [Each byte is made up of 8 bits]

⇒ is given:-

$$\text{Let, } n = 600 \text{ kilobits} \\ = 600 \times 10^3 \text{ bits}$$

$$\text{Required bytes } [n] = \lceil 600 \times 10^3 / 8 \rceil \\ = 75000 \text{ bytes.}$$

Problem: 2

⇒ How many bytes are required to encode 1001 bits of data?

⇒ is given:-

$$\text{let } n = 1001 \text{ bits}$$

we know:-

$$1 \text{ byte} = 8 \text{ bits}$$

So, required bytes:-

$$[n] = \lceil 1001 / 8 \rceil \\ = 126 \text{ bytes.}$$



Problem: 3:-

⇒ In ATM data are organized into cells of 53 bytes.  
How many atm cells can be transmitted in 1 minute over a connection that transmits data at the rate 500 kilobits per second?

⇒ in given:-

$$\text{time } t = 1 \text{ min} = 60 \text{ sec}$$

$$\text{Let, } R = 500 \times 10^3 \text{ bits}$$

$$\begin{aligned} \text{So, In 1 minute this connection can transmit} &= 500 \times 10^3 \times 60 \\ &= 3 \times 10^7 \text{ bits} \end{aligned}$$

we know:-

$$1 \text{ byte} = 8 \text{ bits}$$

$$\therefore 53 \text{ byte} = 53 \times 8 \text{ bits}$$

$$= 424 \text{ bits long.}$$

So, Number of ATM cells that can be transmitted in

$$\begin{aligned} &= \lfloor R/424 \rfloor = \lfloor 3 \times 10^7 / 424 \rfloor \\ &= 70754 \end{aligned}$$

Ans:-

## Lecture 7

### # SET:

⇒ A Set is a collection of well defined, objective.

① we used Capital English letters (A, B, C, D, ...) to denote the sets.

② And used small English letters (a, b, c, d, ...) to denote the elements of sets.

③ Sets can have other sets as members:

Ex:-  $B = \{a, \{a, b\}, \{x\}, y\}$

④ We can represent the set two ways:—

(i) List notation/form:-

⇒ The elements of set are separated by commas and enclosed by curly braces. " $\}$ " " $\{$ "

Ex:-  $A = \{2, 4, 6, 8\}$

(ii) Rule method:-

⇒ A set is defined by specifying a property that elements of the set have in common.

Ex:-  $A = \{x | x \text{ is an even integer between 2 and 8}\}$



## # Types of Sets:-

### Empty set:-

⇒ A set having no element is called as empty set.

It is denoted by  $\emptyset$  or  $\{\}$

Ex:-  $B = \{\}$

$A = \{n \text{ is multiple of } 4 | n \text{ is odd}\}$

Note:-  $\emptyset \neq \{\emptyset\}$

### Singleton set:-

⇒ A set having only one element is called Singleton set.

Ex:-  $A = \{4\}$

$B = \{s\}$

### Finite Set:-

⇒ A set with finite number of element in it, is called a finite set.

Ex:-  $A = \{1, 2, 3, 4\}$

$B = \{a, b, c, d\}$

### Infinite Set:

⇒ An infinite set is a set which contains infinite number of elements.

Ex:-

$$A = \{1, 2, 3, 4, \dots\}$$
$$B = \{1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots\}$$

### Subset:

⇒ A set is called a subset of the set B if for all  $x \in A \Rightarrow x \in B$  /  $\forall x, x \in A \Rightarrow x \in B$ , It is denoted as  $A \subseteq B$

Ex:-

$$B = \{a, e, i, o, u\}$$
$$A = \{a, e, i\}$$

then,  $A \subseteq B$

### Power Set:

⇒ The members of all subsets of set A is called the power set of A. denoted by  $P(A)$

Ex:-

$$A = \{1, 2, 3\}$$

Sub set of A =  $\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}$

$$\therefore P(A) = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}$$



### #Some Examples:-

- ① The set  $V$  of all vowels in the english Alphabet can be written as

$$V = \{a, e, i, o, u\}$$

- ② The set  $O$  of odd positive integers less than 10 can be denote by

$$O = \{1, 3, 5, 7, 9\}$$

- ③ The set of positive integers less than 100 can be denote by  $\{1, 2, 3, 4, \dots, 99\}$

### #Standard Numerical sets:-

- ①  $N = \{0, 1, 2, 3, \dots\} \rightarrow$  Natural numbers.
- ②  $Z = \{\dots, -2, -1, 0, 1, 2, \dots\} \rightarrow$  integers.
- ③  $Z^+ = \{1, 2, 3, \dots\} \rightarrow$  positive integers.
- ④  $Q = \left\{ \frac{p}{q} \mid p \in Z, q \in Z \text{ and } q \neq 0 \right\} \rightarrow$  rational numbers.
- ⑤  $R =$  real numbers  $\rightarrow \left\{ -1, 0, \frac{3}{2}, -\frac{4}{5}, 100, \sqrt{2}, \pi \right\}$

## # $\epsilon$ -Notation:-

$\Rightarrow$  The greek letter " $\epsilon$ " (epsilon) is used to denote that an object is an element of a set. And " $\notin$ " is denote that the object is not an element.

Ex:-  $\exists \epsilon S$  : " $\epsilon$  is an element of the set  $S$ ."

$\exists \notin S$  : " $\epsilon$  is not an element of the set  $S$ ."

Q: Which of the following are true:-

①  $\exists \epsilon \mathbb{R}$   $\longrightarrow$  True

②  $\neg \exists \epsilon \mathbb{N}$   $\longrightarrow$  False

③  $\neg \exists \epsilon \mathbb{R}$   $\longrightarrow$  True

④  $0 \notin \mathbb{Z}^+$   $\longrightarrow$  True

⑤  $\exists n, n \in \mathbb{R} \wedge n^2 = -5$   $\longrightarrow$  False



## #Venn Diagram:-

⇒ Set can be represented graphically using Venn diagram.

① In Venn diagrams, the Universal set  $U$ , which contains all the objects under consideration, it is represented by a rectangle.

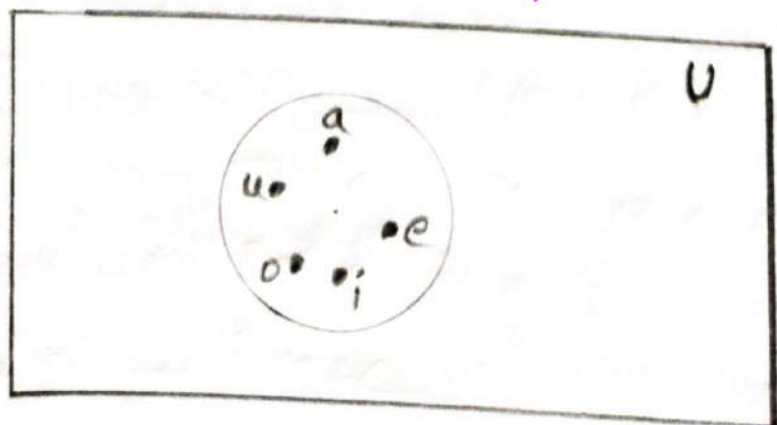
② Inside the rectangle,

(i) Circles or other geometrical figures are used to represent sets.

(ii) Sometime points are used to represent the particular element of the set.

③ Venn Diagrams are often used to indicate the relationships between sets.

✶ Venn Diagram for the set of Vowels:-



## # Equal Sets —

⇒ Two Sets are equal if and only if they have the same elements.

$$A=B \text{ iff } \forall x (x \in A \leftrightarrow x \in B)$$

①.  $A \subseteq B$  and  $B \subseteq A$  then it is written  $A=B$

## Example —

⇒ The set  $A = \{1, 3, 5\}$  and  $B = \{3, 5, 1\}$  are equal because they have the same elements.

## # Subset —

⇒ The set  $A$  is a subset of  $B$  if and only if every element of  $A$  is also an element of  $B$ .

①.  $A \subseteq B$  to indicate that  $A$  is a subset of  $B$

②.  $\forall x (x \in A \rightarrow x \in B) \rightarrow$  is true for  $A \subseteq B$

③. (i) Every non-empty set  $S$  is guaranteed to have at least two subsets.

(ii) The empty set and the set  $S$  itself, that is  $\emptyset \subseteq S$  and  $S \subseteq S$



### # Proper Subset! —

⇒ Any Subset  $A$  is said to be proper subset of another set  $B$ . If  $A$  is a subset of  $B$ , but there is at least one element of  $B$  which does not belong to  $A$ .

① if  $A \subseteq B$  but  $A \neq B$

②  $\forall x (x \in A \rightarrow x \in B) \wedge \exists x (x \in B \wedge x \notin A)$

③ we denote  $A \subset B = A$  is a proper set of  $B$ .

Ex:-

$$A = \{1, 5\}, B = \{1, 5, 6\}$$

⇒ Here,  $A$  is a proper set of  $B$   
 $A \subset B$

### # Subset Example! —

Q: Which of the following are true? —

1.  $\mathbb{Z} \subseteq \mathbb{N} \rightarrow \text{False}$

2.  $\mathbb{N} \subseteq \mathbb{R} \rightarrow \text{True}$

3.  $-\mathbb{Z} \subseteq \mathbb{R} \rightarrow \text{False}$

4.  $\{1, 2\} \notin \mathbb{Z}^+ \rightarrow \text{False}$

5.  $\emptyset \subseteq \emptyset \rightarrow \text{True}$

6.  $\emptyset \subset \emptyset \rightarrow \text{False}$

### #Quick examples:

- ①  $\{1, 2, 3\} \subseteq \{1, 2, 3, 4, 5\} \rightarrow \text{True}$
- ②  $\{1, 2, 3\} \subset \{1, 2, 3, 4, 5\} \rightarrow \text{True}$
- ③ in  $\emptyset \subseteq \{1, 2, 3\} ? \rightarrow \text{yes}$
- ④ in  $\emptyset \in \{1, 2, 3\} ? \rightarrow \text{NO}$
- ⑤ in  $\emptyset \subseteq \{\emptyset, 1, 2, 3\} ? \rightarrow \text{yes}$
- ⑥ in  $\emptyset \in \{\emptyset, 1, 2, 3\} ? \rightarrow \text{yes}$
- ⑦ in  $\{x\} \subseteq \{x\} ? \rightarrow \text{yes}$
- ⑧ in  $\{x\} \in \{x, \{x\}\} ? \rightarrow \text{yes}$
- ⑨ in  $\{x\} \subset \{x, \{x\}\} ? \rightarrow \text{yes}$



## # Cardinality of a Set:-

$\Rightarrow$  If there are exactly  $n$  distinct members in the set  $S$ , we say that  $S$  is a finite set and  $n$  is cardinality of  $S$ .

- ① The cardinality of a set is the number of distinct elements in the set.
- ②  $|S|$  denotes by cardinality of set  $S$ .

Example:-

①  $A = \{ \}$

$|A| = 0$

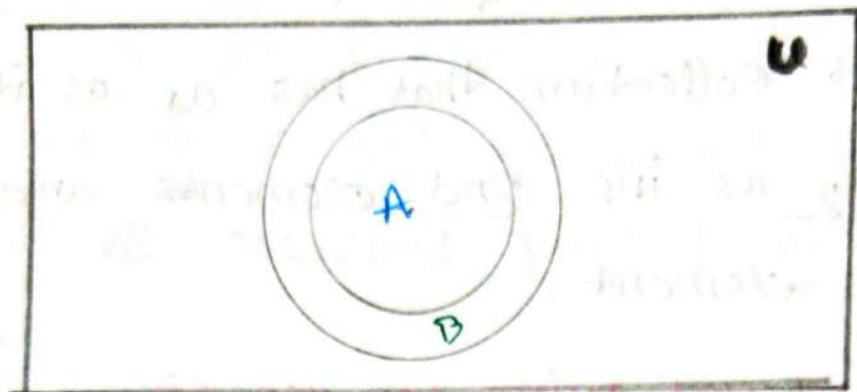
②  $A = \{1, 2, 3, 4\}$

$|A| = 4$

Q:- Compute the cardinality of each of the sets?

- ①  $A = \{1, -1, 4, -1, 1\} \Rightarrow |A| = 3$  or  $A = \{1, 4, -1\}$
- ②  $A = \{3, \{3, 4, 5\}, \emptyset\} \Rightarrow |A| = 3$
- ③  $A = \{\}$   $\Rightarrow |A| = 0$
- ④  $A = \{\{\}, \{\}, \{\}\} \Rightarrow |A| = 3$
- ⑤  $A = \{1, 2, 3, \dots\} \Rightarrow |A| = \text{infinite}$

## # Venn Diagram of $A \subseteq B$



## # Super Set!

$\Rightarrow$  If  $A$  is a subset of  $B$ , then  $B$  is called Super set of  $A$ . Denote by  $B \supseteq A$  and read  $B$  is super set of  $A$ .

①  $A \supseteq V = A$  is a Super set of  $V$

Ex:  $A = \{1, 2, 7, 8\}$ ,  $B = \{4, 7\}$

So,  $A$  is a Super set of  $B$  or  $A \supseteq B$



## # Ordered n-tuples:

$\Rightarrow$  The Ordered n-tuples  $(a_1, a_2, \dots, a_n)$  is the ordered collection that has  $a_1$  as its first element,  $a_2$  as its 2nd elements and  $a_n$  as its nth element.

① For Ordered pairs, we use  $()$

②  $(11, 11, 11, 12, 13) \neq (11, 12, 13)$  But  
 $\{11, 11, 11, 12, 13\} = \{11, 12, 13\}$

③ Two Ordered n-tuples are equal when,

Let  $A = (a_1, a_2, a_3, \dots, a_n)$   
 $B = (b_1, b_2, b_3, \dots, b_n)$

$\Rightarrow$  if and only if  $a_1 = b_1$   
 $a_2 = b_2$   
 $a_n = b_n$

④ Two - Tuples are called ordered pairs

Let  $A = (a, b)$   
 $B = (c, d)$

$\Rightarrow$  if and only if  $a = c$   
 $b = d$

Note -  $(a, b)$  and  $(b, a)$  are not equal unless  $a = b$

## # Cartesian Product of Sets:

⇒ Let, A and B be sets, The Cartesian product of A and B denoted by  $A \times B$ , is the set of all ordered pairs (a, b) where  $a \in A$  and  $b \in B$

$$1. A \times B = \{(a, b) \mid a \in A \text{ and } b \in B\}$$

Example:-

⇒ What is the Cartesian product of  $A = \{1, 2\}$  and  $B = \{a, b, c\}$ ?

⇒ is given:-

$$A = \{1, 2\}$$

$$B = \{a, b, c\}$$

we know:-

$$A \times B = \{1, 2\} \times \{a, b, c\}$$

$$= \{(1, a), (1, b), (1, c), (2, a), (2, b), (2, c)\}$$

2.  $A \times B$  and  $B \times A$  are not equal

3. At least 2 sets are needed to find the Cartesian product.

4. A subset of the Cartesian product  $A \times B$  is called the Set A to the Set B Partial form.



Example:-

$\Rightarrow$  if  $A = \{1, 2\}$ ,  $B = \{3, 4\}$ ,  $C = \{5, 6, 7\}$  what is the  $A \times B \times C$ ?

$\Rightarrow$  in given:-

$$A = \{1, 2\}$$

$$B = \{3, 4\}$$

$$C = \{5, 6, 7\}$$

we know:-

$$A \times B \times C = \{(1, 3, 5), (1, 3, 6), (1, 3, 7), (1, 4, 5), (1, 4, 6), (1, 4, 7), \\ (2, 3, 5), (2, 3, 6), (2, 3, 7), (2, 4, 5), (2, 4, 6), (2, 4, 7)\}$$

note:-  $|A \times B \times C| = |A| \cdot |B| \cdot |C|$

$$= 2 \times 2 \times 3$$

$$= 12$$

## Lecture 8:-

### # Union of Sets:-

⇒ Let, A and B the sets, the Union of the set denote by  $A \cup B$ . in the set that contains those element that are either in A or in B or in both.

$$\textcircled{1} \quad A \cup B = \{x | x \in A \vee x \in B\}$$

Example 1:-

⇒ What is the Union of the set  $A = \{1, 3, 5\}$  and  $B = \{1, 2, 3\}$

⇒ in given:-

$$A = \{1, 3, 5\}$$

$$B = \{1, 2, 3\}$$

$$\begin{aligned} A \cup B &= \{1, 2, 3\} \cup \{1, 3, 5\} \\ &= \{1, 2, 3, 5\} \end{aligned}$$

Example 2:-

⇒ if  $A = \{\text{Moon, Farhan, Alfe, Soheli}\}$

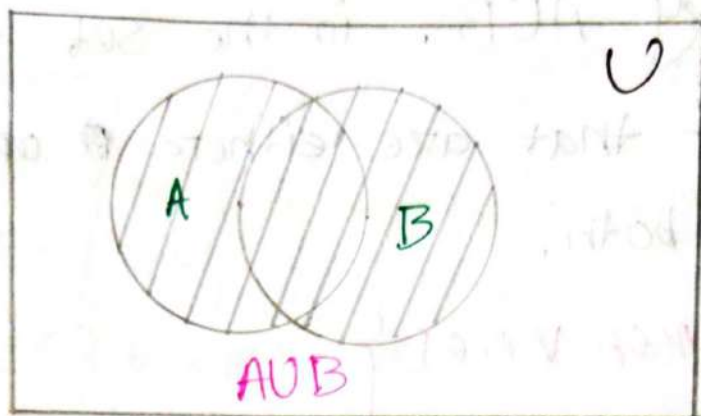
$B = \{\text{Jam, Araf, Baw, Sami, Mohsin}\}$

What is the  $A \cup B$ ?

⇒  $\therefore A \cup B = \{\text{Moon, Farhan, Alfe, Soheli, Jam, Araf, Baw, Sami, Mohsin}\}$



## # Venn Diagram of $A \cup B$



$$A \cup B = \{x | x \in A \vee x \in B\}$$

## # Intersection of sets:

$\Rightarrow$  Let, A and B be sets, The intersection of the sets A and B denote by  $A \cap B$  in the containing elements in both A and B

$$① \quad A \cap B = \{x | x \in A \wedge x \in B\}$$

Example:-

$\Rightarrow$  what is the intersection of sets  $A = \{1, 2, 3\}$   
 $B = \{3, 4, 5\}$ ?

$\Rightarrow$  is given:

$$A = \{1, 2, 3\}, B = \{3, 4, 5\}$$

$$\therefore A \cap B = \{3\}$$

Example 4: —

if  $B = \{x | x \text{ is an odd positive integer and } x \leq 5\}$

$C = \{x | x \text{ is prime number less than } 10\}$

What is  $B \cap C$ ?

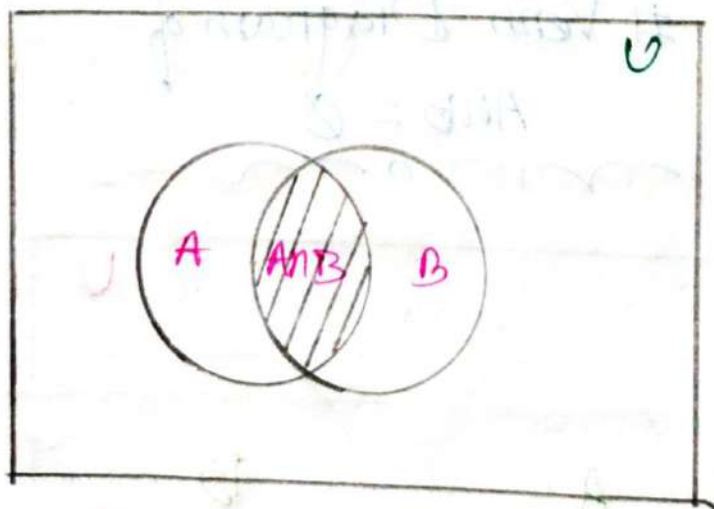
⇒ in given: —

$$B = \{1, 3, 5\}$$

$$C = \{2, 3, 5, 7\}$$

$$\begin{aligned}\therefore B \cap C &= \{1, 3, 5\} \cap \{2, 3, 5, 7\} \\ &= \{3, 5\}\end{aligned}$$

# Venn Diagram of  $A \cap B$



$$A \cap B = \{x | x \in A \wedge x \in B\}$$



## # Disjoint set:-

⇒ Two sets are disjoint if their intersection is the empty set.

① If  $A \cap B = \emptyset$  it's a disjoint set

② Disjoint = Non-intersecting

Example: 5

⇒ Let  $A = \{1, 2, 3\}$ ,  $B = \{4, 5, 6\}$  what is the  $A \cap B$

⇒ is given:-

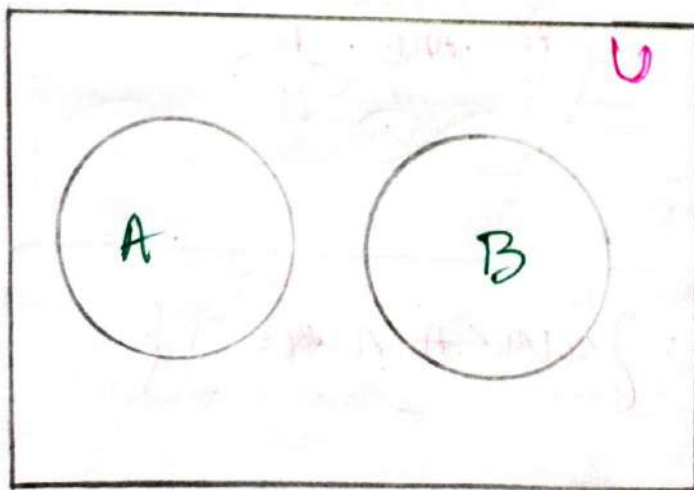
$$A = \{1, 2, 3\}$$

$$B = \{4, 5, 6\}$$

∴  $A \cap B = \{\}$  → A and B Disjoint

# Venn Diagram of

$$A \cap B = \emptyset$$



## # Principle of inclusion-exclusion:-

Q:- How many elements are there in the union of two sets A and B? or what is the Cardinality of a union of two sets A and B.

⇒ The number of elements in the Union of two is the sum of the number of distinct element in the set. or

The sum of the number of element in the set minus the number of element in there intersection.

$$|A \cup B| = |A| + |B| - |A \cap B|$$

① The generalization of this result to unions of the arbitrary number of sets is called Principle of inclusion - exclusion.



## # Difference of Two Sets:-

$\Rightarrow$  The difference of the set A and B denoted by  $A-B$  is the set containing those element that are in A but not in B.

①  $A-B = \{x | x \in A \wedge x \notin B\}$

② The Difference  $A-B$  is called the complement of B with respect to A.

$$A-B = A \cap \bar{B}$$

Ex:-6:-

Let, The Difference  $A = \{1, 3, 5\}$  and  $B = \{1, 2, 3\}$  in the set. What is the Difference  $A-B$  and  $B-A$ ?

$\Rightarrow$  in given:-

$$A = \{1, 3, 5\}$$

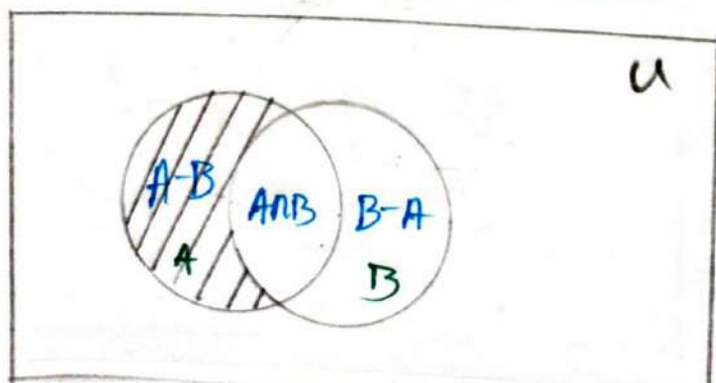
$$B = \{1, 2, 3\}$$

$$\therefore A-B = \{5\}$$

$$B-A = \{2\}$$

# Venn Diagram of

$A-B$



## # Complement of Set:-

$\Rightarrow$  Let  $U$  is the Universal set. The complement of set  $A$ , denote by  $\bar{A}$ . is the complement of  $A$  with respect to  $U$ .

①  $\bar{A} = \{x | x \notin A\}$

② In other words, the complement of set  $A$  is  $U - A$ ,

$$\therefore \bar{A} = U - A$$

Ex:-  $U = \{1, 2, 3, 4, 5, 6, 7\}$  and  $A = \{1, 4, 5\}$ . find the Complement of set  $A$

$\Rightarrow$  in given:-  $U = \{1, 2, 3, 4, 5, 6, 7\}$

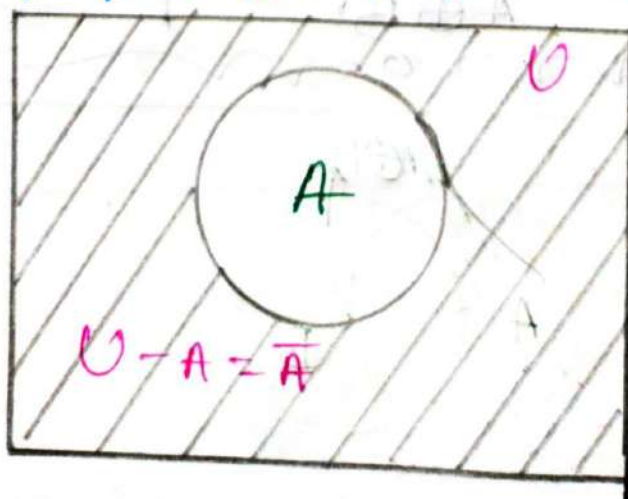
$$A = \{1, 4, 5\}$$

$$\therefore \bar{A} = U - A$$

$$= \{2, 3, 6, 7\}$$

# Venn Diagram of the

$U - A$





## # Symmetric Difference of

Two sets:

⇒ The Symmetric Difference of two sets  $A, B$  denote  $A \oplus B$ , is the set of elements that belongs to  $A$  or to  $B$ . But not the Both  $A$  and  $B$

①  $A \oplus B = \{x | (x \in A \wedge x \notin B) \vee (x \in B \wedge x \notin A)\}$

②  $A \oplus B = (A - B) \cup (B - A)$

Example:—

⇒ If  $A = \{1, 2, 3\}$ ,  $B = \{2, 4, 6\}$  what is the symmetric Difference of 2 set

⇒ in given:—

$$A = \{1, 2, 3\}$$

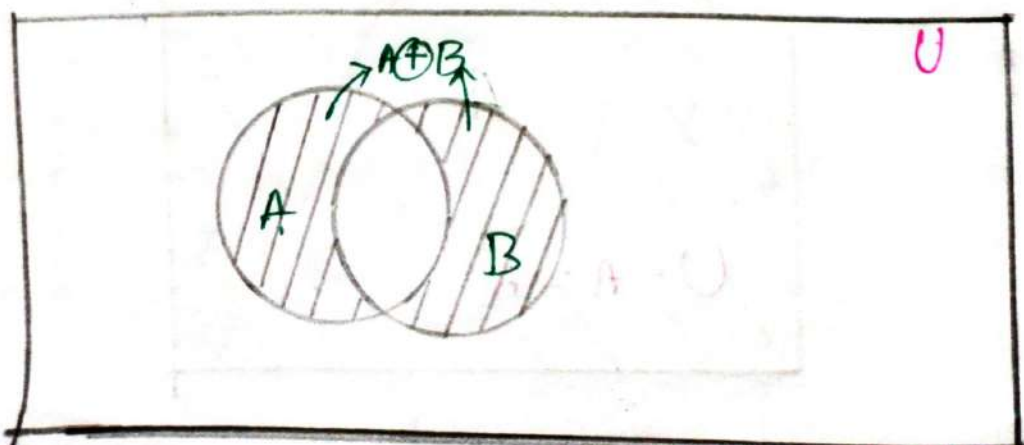
$$B = \{2, 4, 6\}$$

$$A \oplus B = (A - B) \cup (B - A)$$

$$= \{1, 3, 4, 6\}$$

# Venn Diagram of

$$A \oplus B$$



# #Set Identities: —

Identity

Name

1.  $A \cup \emptyset = A$

2.  $A \cap U = A$

Identity Law

3.  $A \cup U = U$

4.  $A \cap \emptyset = \emptyset$

Domination Law

5.  $A \cup A = A$

6.  $A \cap A = A$

Idempotent Law

8.  $\overline{\overline{A}} = A$

Complementation Law

9.  $A \cup B = B \cup A$

10.  $A \cap B = B \cap A$

Commutative Law

11.  $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

12.  $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

Associative Laws

13.  $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

14.  $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

Distributive Laws

15.  $\overline{A \cup B} = \overline{A} \cap \overline{B}$

16.  $\overline{A \cap B} = \overline{A} \cup \overline{B}$

De - Morgans Laws

17.  $A \cup (A \cap B) = A$

18.  $A \cap (A \cup B) = A$

Absorption Laws

19.  $A \cup \overline{A} = U$

20.  $A \cap \overline{A} = \emptyset$

Complement Laws.



## # Membership Table:

⇒ A table displaying the membership of elements of sets.

Problem: (10)

Using membership table show that:-

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

| A | B | C | $B \cup C$ | $A \cap B$ | $A \cap C$ | $A \cap (B \cup C)$ | $(A \cap B) \cup (A \cap C)$ |
|---|---|---|------------|------------|------------|---------------------|------------------------------|
| 1 | 1 | 1 | 1          | 1          | 1          | 1                   | 1                            |
| 1 | 1 | 0 | 1          | 1          | 0          | 1                   | 1                            |
| 1 | 0 | 1 | 1          | 0          | 1          | 1                   | 1                            |
| 1 | 0 | 0 | 0          | 0          | 0          | 0                   | 0                            |
| 0 | 1 | 1 | 1          | 0          | 0          | 0                   | 0                            |
| 0 | 1 | 0 | 1          | 0          | 0          | 0                   | 0                            |
| 0 | 0 | 1 | 1          | 0          | 0          | 0                   | 0                            |
| 0 | 0 | 0 | 0          | 0          | 0          | 0                   | 0                            |

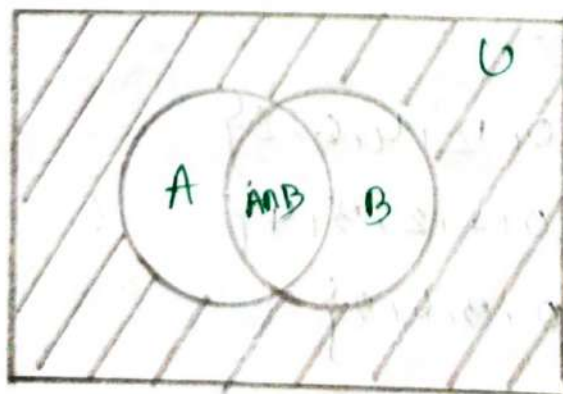
Q Proved that  $(\overline{A \cup B}) = \bar{A} \cap \bar{B}$  using membership table.

| A | B | $\bar{A}$ | $\bar{B}$ | $A \cup B$ | $\overline{A \cup B}$ | $\bar{A} \cap \bar{B}$ |
|---|---|-----------|-----------|------------|-----------------------|------------------------|
| 0 | 0 | 1         | 1         | 0          | 1                     | 1                      |
| 0 | 1 | 1         | 0         | 1          | 0                     | 0                      |
| 1 | 0 | 0         | 1         | 1          | 0                     | 0                      |
| 1 | 1 | 0         | 0         | 1          | 0                     | 0                      |

Since the corresponding values of  $\overline{A \cup B}$  and  $\bar{A} \cap \bar{B}$  are same so  $\overline{A \cup B} = \bar{A} \cap \bar{B}$

Proved

# Venn Diagram of  $\overline{A \cup B}$  and  $\bar{A} \cap \bar{B}$





Q:- Prove that  $\overline{(A \cup B)} = \bar{A} \cap \bar{B}$  using set builder notation and logically equivalences.



$$\begin{aligned}\overline{(A \cup B)} &= \{x \mid \neg(x \in A \vee x \in B)\} \\ &= \{x \mid \neg(x \in A) \wedge \neg(x \in B)\} \\ &= \{x \mid (x \in \bar{A}) \wedge (x \in \bar{B})\} \\ &= \bar{A} \cap \bar{B}\end{aligned}$$

# Union and intersection of three sets:-

Let,  $A = \{0, 2, 4, 6, 8\}$

$$B = \{0, 1, 2, 3, 4\}$$

$$C = \{0, 3, 6, 9\}$$

What are  $A \cup B \cup C$  and  $A \cap B \cap C$ ?

Ans:- in given:-

$$A = \{0, 2, 4, 6, 8\}$$

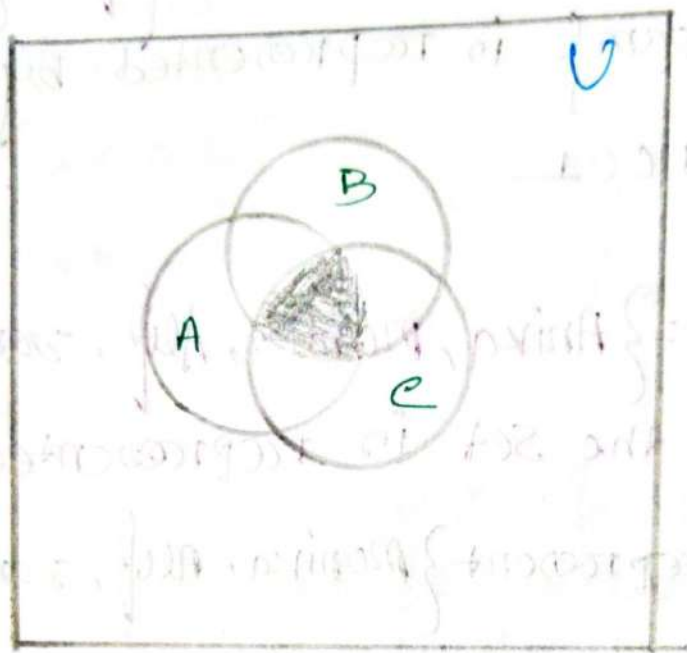
$$B = \{0, 1, 2, 3, 4\}$$

$$C = \{0, 3, 6, 9\}$$

$$A \cup B \cup C = \{0, 1, 2, 3, 4, 6, 8, 9\}$$

$$A \cap B \cap C = \{0\}$$

## # Venn Diagram of $A \cap B \cap C$



## # Computer Representation of Sets

⇒ If we Order the elements of our Universe we can represent by bit-string. for example consider the Universe.

$U = \{ \text{Anika, monika, Alef, Jam} \}$

- ① Order the element alphabetically
- ② Subset of U are represent by bit-string of length 4.
- ③ Each bit in turn, tells us whether the corresponding element is contained in the set.



Question: 1

$U = \{ \text{Anika, Monika, Alif, Jam} \}$

What are the bit-string for  $\{ \text{Anika, Jam} \}$

$\Rightarrow \{ \text{Anika, Jam} \}$  is represented by the bit-string 1001

Q: 2:  $U = \{ \text{Anika, Monika, Alif, Jam} \}$

What the set is represented by 0111

$\Rightarrow 0111$  represent  $\{ \text{Monika, Alif, Jam} \}$

Q: 3 Let  $U = \{ \text{Jam, Mohsin, Farrah, Alif} \}$

What is the symmetric difference of  $\{ \text{Farrah} \}$  with  $\{ \text{Jam, Farrah, Alif} \}$  in CRS

$\Rightarrow$  In given:

$U = \{ \text{Jam, Mohsin, Farrah, Alif} \}$

Let  $A = \{ \text{Farrah} \}$

$B = \{ \text{Jam, Farrah, Alif} \}$

In CRS

$A = 0010$

$B = 1011$

$U = 1111$

So, the Symmetric Difference of A with B  
is represented by:

$$\begin{array}{r} 0010 \\ \oplus 1011 \\ \hline 1001 \end{array}$$

$\Rightarrow \{ \text{sym. Diff} \}$

Question 4:

Let  $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$

- What bit string represents the subset of all odd integers in  $U$ ?
- What bit string represents the subset of even integers of  $U$ ?
- What bit string represents the subsets of integers not exceeding 5 in  $U$ ?

$\Rightarrow$

Let, Odd  $A = \{1, 3, 5, 7, 9\}$

$\therefore \{1, 3, 5, 7, 9\} \Rightarrow 1010101010$

$\{2, 4, 6, 8, 10\} \Rightarrow 0101010101$

$\{1, 2, 3, 4, 5\} \Rightarrow 1111100000$



Question: 6:-

⇒ What is the bit string for the complement of the set  $\{1, 3, 5, 7, 9\}$  where  $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$

⇒ In given:-

$$U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$$

$$A = \{1, 3, 5, 7, 9\}$$

We know:-

$$\text{Complement of set } \bar{A} = U - A$$

$$= \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\} - \{1, 3, 5, 7, 9\}$$

$$= \{2, 4, 6, 8, 10\}$$

So, The bit string of  $\bar{A}$  Set is  $= 0101010101$

Question 6:-

⇒ The bit string of set  $\{1, 2, 3, 4, 5\}$  and  $\{1, 3, 5, 7, 9\}$  are 1111100000 and 1010101010, respectively.

Use bit string to Find the Union and Intersection of these sets:-?

⇒ In given:-

Let,  $A = \{1, 2, 3, 4, 5\} \Rightarrow 1111100000$

$B = \{1, 3, 5, 7, 9\} \Rightarrow 1010101010$

Find Union:-

$$\begin{array}{r} 1111100000 \\ (+) 1010101010 \\ \hline 1111101010 \end{array}$$

So, the  $A \cup B = \{1, 2, 3, 4, 5, 7, 9\}$

Find the Intersection:-

$$\begin{array}{r} 1111100000 \\ (x) 1010101010 \\ \hline 1010100000 \end{array}$$

So, the set of

$A \cap B = \{1, 3, 5\}$



## Lecture 9!

### # The Integers and Division!

⇒ if  $a$  and  $b$  are the integers with  $a \neq 0$  we can say that  $a$  divides  $b$  if there is an integer  $c$  such that  $b = ac$ . when  $a$  divides  $b$  we say

(1)  $a$  is a factor of  $b$ .

(2)  $b$  is a multiple of  $a$ .

(i) The notation  $a|b$  denotes that  $a$  divides  $b$ .

(ii) we write  $a \nmid b$  when  $a$  does not divide  $b$ .

#### Problem 1

⇒ Determine whether  $3|7$  and whether  $3|12$ .

⇒ To Determine  $3|7$  and  $3|12$ , we need to check if  $7$  or  $12$  is divisible by  $3$  without leaving a remainder.

(1)  $3$  does not  $| 7$  Because  $7/3$  is not integer.

(2)  $3|12$  Because  $12/3$  is a integer  $12/3 = 4$ .

Q: Which the following is true?

- ①  $77|7 \rightarrow \text{False} \rightarrow$  bigger number can't divide smaller positive number
- ②  $7|77 \rightarrow \text{True} \rightarrow 77/7 = 11 \rightarrow b = a \cdot c \rightarrow 77 = 7 \cdot 11$
- ③  $24|24 \rightarrow \text{True} \rightarrow 24/24 = 1 \rightarrow b = a \cdot c \rightarrow 24 = 24 \cdot 1$
- ④  $0|24 \rightarrow \text{False} \rightarrow 24/0 = \infty$
- ⑤  $24|0 \rightarrow \text{True} \rightarrow 0/24 = 0 \rightarrow b = a \cdot c \rightarrow 0 = 24 \cdot 0$

### # Some Basic properties of Divisibility of integers:

$\Rightarrow$  Let  $a, b, c$  be integers, then

- i) if  $a|b$  and  $a|c$  then  ~~$a|b+c$~~   $a|(b+c)$
- ii) if  $a|b$ , then  $a|b+ce$  all integers for  $e$
- iii) if  $a|b$  and  $b|c$  then  $a|c$
- iv) if  $a|b$  and  $a|c$  then

$$a(mb+nc)$$

$m, n = \text{integers}$



Example: 2  
non

- (i) if  $17|34$  and  $17|120$  then  $17|120+34$
- (ii) if  $17|34$  then  $17|34 \times 3$  or  $17|34 \times 4$
- (iii) if  $17|34$  and  $34|120$  then  $17|120$
- (iv) if  $17|34 \cdot 10$  and  $17|120 \times 5$  then  $17|34 \cdot 10 + 120 \times 5$

# Division:

$$a = dq + r$$

$d = \text{divisor}$   $\leftarrow 31$   $\xrightarrow{3} q = \text{quotient}$

$$\begin{array}{r} 31 \overline{) 117} \\ \underline{93} \phantom{00} \\ 24 \end{array}$$

$\rightarrow \text{dividend} = 117$

$r = \text{remainder}$

$$117 = 31 \cdot 3 + 24$$

## #The Division Algorithm:—

①  $a = dq + r$  ;  $0 \leq r < d$

$d = \text{divisor}$

$a = \text{divide no}$

$q = \text{quotient}$

$r = \text{remainder}$

② The notation is used to express the quotient and remainder.

$$q = a \text{ div } d$$

$$r = a \text{ mod } d$$

Example: →

⇒ What are the quotient and remainder when 101 is divided by 11?

⇒ is given,

$$a = 101$$

$$d = 11$$

$$\text{So, } 101 = 11 \cdot 9 + 2$$

$$9 = 101 \text{ div } 11$$

$$2 = 101 \text{ mod } 11$$



Compute —

①  $113 \bmod 24$

⇒ here,

$$a = 113$$

$$d = 24$$

$$q = 4$$

$$r = 17$$

$$\begin{array}{r} 4 \\ 24 \overline{) 113} \\ \underline{96} \\ 17 \end{array}$$

$$\therefore 17 = 113 \bmod 24$$

②  $-20 \bmod 7$

here,  $a = -20$

$$d = 7$$

$$q = -5$$

$$r = 6$$

$$\begin{array}{r} -5 \\ 7 \overline{) -20} \\ \underline{-35} \\ 6 \end{array}$$

$$\therefore 6 = -20 \bmod 7$$

Example 4:

⇒ what are the quotient and remainder when  $-11$  is divided by 3

⇒ is given: —

$$\begin{array}{l} a = -11 \\ d = 3 \end{array}$$

we know —

$$a = qd + r$$

$$-11 = (-4) \cdot 3 + 1$$

So, the quotient when  $-11$  is divided by  $3$

$$-4 = -11 \div 3$$

and, the remainder is

$$1 = -11 \bmod 3$$

### # Congruence:

$\Rightarrow$  If  $a$  and  $b$  are the integers and  $m$  is a positive integer, then  $a$  is congruent to  $b$  modulo  $(m)$  if  $m \mid (a-b)$

① Defines  $a \equiv b \pmod{m}$

② if  $a$  and  $b$  not congruent modulo  $m$ , we can write,  $a \not\equiv b \pmod{m}$

③  $a \equiv b \pmod{m}$  iff  $a \bmod m = b \bmod m$



Example 8:—

⇒ Determine whether 17 is congruent to 5 modulo 6 and whether the 24 and 14 are congruent modulo 6.

⇒ is given:—

$$\text{Let } a = 17$$

$$b = 5$$

$$\text{Since, } 6 \mid (17 - 5) \Rightarrow 6 \mid 12$$

$$\text{So, } 17 \equiv 5 \pmod{6}$$

Again:—

$$\text{Let, } a' = 24$$

$$b' = 14$$

$$\text{Since, } 6 \nmid (24 - 14) \Rightarrow 6 \nmid 10$$

$$\text{So, } 24 \not\equiv 14 \pmod{6}$$

Question: — Which of the following are true?

- ①  $3 \equiv 3 \pmod{17} \rightarrow \text{True}$   $17 | 3-3 \Rightarrow 0/17 = 0$
- ②  $3 \equiv -3 \pmod{17} \rightarrow \text{False}$   $17 \nmid 3+3 \Rightarrow 6/17 = \text{fraction value}$
- ③  $172 \equiv 177 \pmod{5} \rightarrow \text{True}$   $5 | -5 \Rightarrow -5/5 = -1$
- ④  $-13 \equiv 13 \pmod{26} \rightarrow \text{True}$   $-13 | 3-13 \Rightarrow -26/-13 = 2$

Question: —

① List 5 integers that are congruent to 4 modulo 12.

$\Rightarrow$  ①  $16 \equiv 4 \pmod{12}$

②  $28 \equiv 4 \pmod{12}$

③  $40 \equiv 4 \pmod{12}$

④  $52 \equiv 4 \pmod{12}$

⑤  $64 \equiv 4 \pmod{12}$



② List 5 integers that are congruent to 3 modulo 7.

⇒ (i)  $10 \equiv 3 \pmod{7}$

(ii)  $17 \equiv 3 \pmod{7}$

(iii)  $24 \equiv 3 \pmod{7}$

(iv)  $31 \equiv 3 \pmod{7}$

(v)  $38 \equiv 3 \pmod{7}$

③ List 5 integers that are congruent to 5 modulo 17.

⇒ (i)  $22 \equiv 5 \pmod{17}$

(ii)  $39 \equiv 5 \pmod{17}$

(iv)  $56 \equiv 5 \pmod{17}$

(iii)  $73 \equiv 5 \pmod{17}$

(v)  $90 \equiv 5 \pmod{17}$

## # Cryptology :-

- ① Cryptology is the study of secret message.
- ② Julius Caesar made this.
- ③ He made messages by shifting each letter three letters forward in the alphabet.

Ex:- B  $\rightarrow$  E  
X  $\rightarrow$  A

④

## # Caesar Cipher:-

Mathematical expression:-

$$f(p) = (p+3) \bmod 26 \quad 0 \leq p \leq 25$$

- ① There are 26 letters in English alphabet.  
So, A=0, B=1, ..., K=10, ..., Z=25



# # Letter $\leftrightarrow$ Number Conversion Table:

|   |   |   |   |   |   |   |   |   |   |    |    |    |
|---|---|---|---|---|---|---|---|---|---|----|----|----|
| A | B | C | D | E | F | G | H | I | J | K  | L  | M  |
| 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 |

|    |    |    |    |    |    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|----|----|----|----|----|
| N  | O  | P  | Q  | R  | S  | T  | U  | V  | W  | X  | Y  | Z  |
| 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 |

Problem: 7

⇒ what is the secret message produced from the message "MEET you in the PARK" using Caesar cipher.

⇒ in given: —

the message is = "MEET YOU IN THE PARK"

⇒ 1st replace the letters in the message with number

This produces: 12 4 4 19 24 14 20 8 13 10 7 4  
15 0 17 10

⇒ Now, replace these numbers  $\cdot P$  by  $f(P) = (P+3) \bmod 26$ . This produces.

15 7 7 22 1 17 23 11 16 22 10 7  
18 13 20 13

⇒ Translate this back to letters for produces the encrypted message

"PHHW BRX LQ SDVN"

### # Decryption:

1. To recover the original message from a secret message by the Caesar cipher, the function  $f^{-1}$  is used.

2.  $f^{-1}(P) = (P-3) \bmod 26$

3. The process of determining the original message from the encrypted message is called decryption.



Problem: 8! —

⇒ Decrypt the encrypted message "PHHW BRX  
LQ WKH SDON"

⇒ 1st replace the letters in the message with number

15 7 7 22 11 7 23 11 16 22 10 7 18 3 20 13

⇒ now, Replace each of these number  $p$  by  $f^{-1}(p) = (p-3) \bmod 26$  this produces:

12 4 4 19 24 14 20 8 13 19 7 4 15 0 17 10

⇒ Translating this back to letters produce the  
Decrypt message:

"MEET YOU IN THE PARK"

## #Shift cipher:-

⇒ If we can shift each letter by  $k$ , so that

$$\textcircled{1} f(p) = (p+k) \bmod 26$$

Such a cipher is called a shift cipher

$\textcircled{2}$  The decryption can be carried out using

$$f^{-1}(p) = (p-k) \bmod 26$$

Problem:- 10!

Encrypt the message "DO NOT PASS GO" by translating the letters into numbers, Applying the encryption function given, and then translating the number back into letters.

$$f(p) = (p+13) \bmod 26$$

⇒ The message in numbers:-

3, 14

10, 14, 10

15, 0, 18, 18

6, 14

⇒ Replace the each numbers  $p$  by  $f(p) = (p+13) \bmod 26$  produces:-

16, 1

0, 1, 6

2, 10, 5, 5,

17, 1



⇒ Translating this back into letters produces the encrypted message:—

“QB ABG CNFF TB”

Problem:—14:—

⇒ Decrypt the encrypted message “QB. ABG CNFF TB” which is encrypted using  $f(p) = (p + 13) \bmod 26$

⇒ The encrypted message in number:—

16, 1      0, 1, 6      2, 13, 15, 15      10, 1

⇒ Replace each Number  $p$  by  $f^{-1}(p) = (p - 13) \bmod 26$

3, 14      13, 14, 10      15, 0, 18, 18      6, 14

⇒ Translating this back to the letters produces the Decrypt the encrypted message:—

“DO NOT PASS GO”

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**End**

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